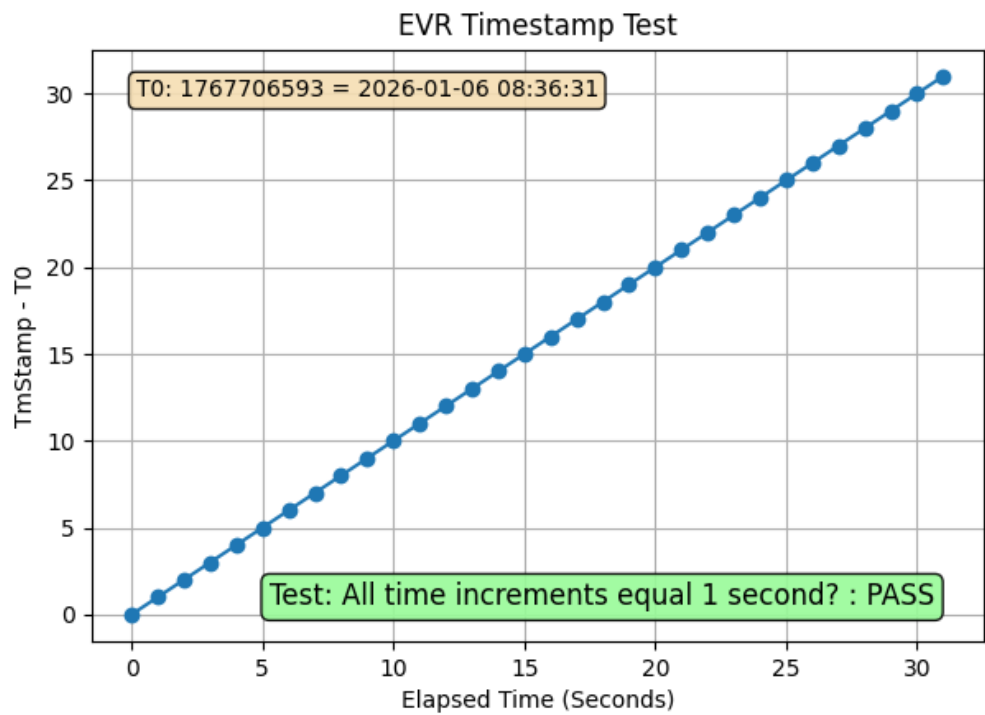


PSC Functional Test Report

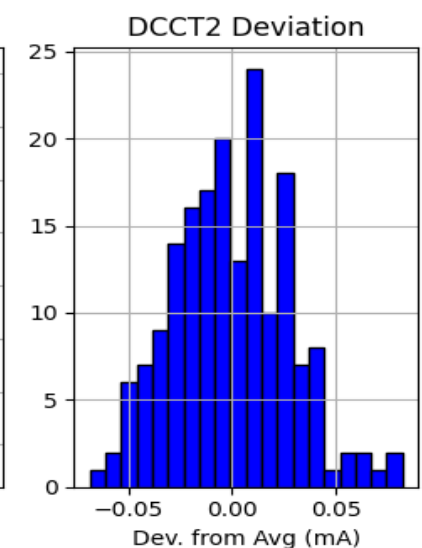
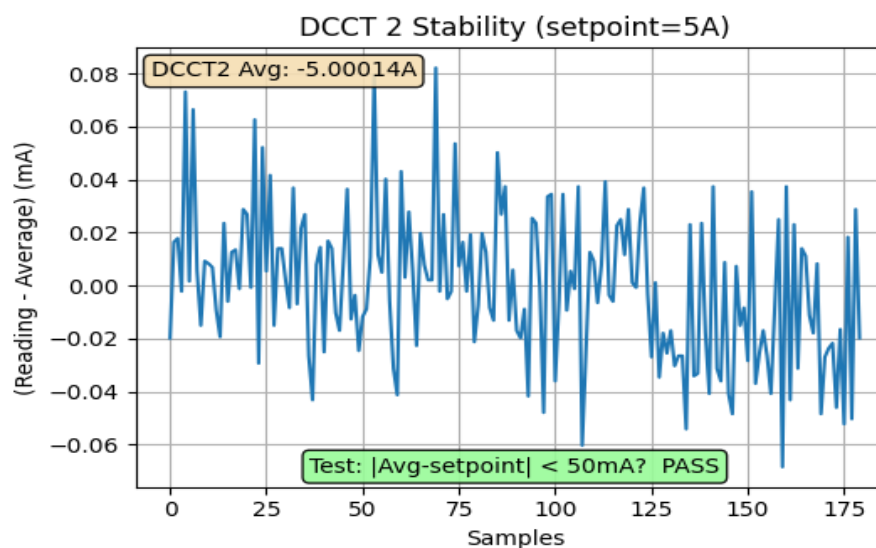
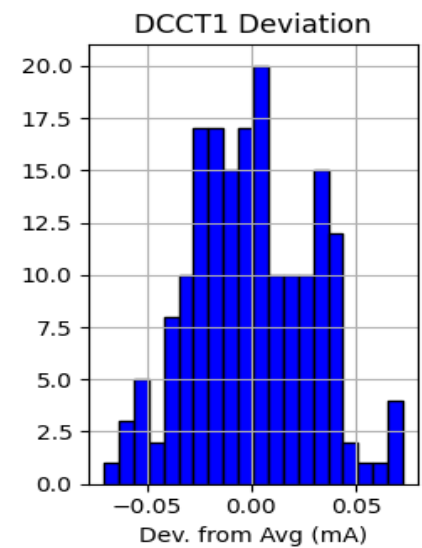
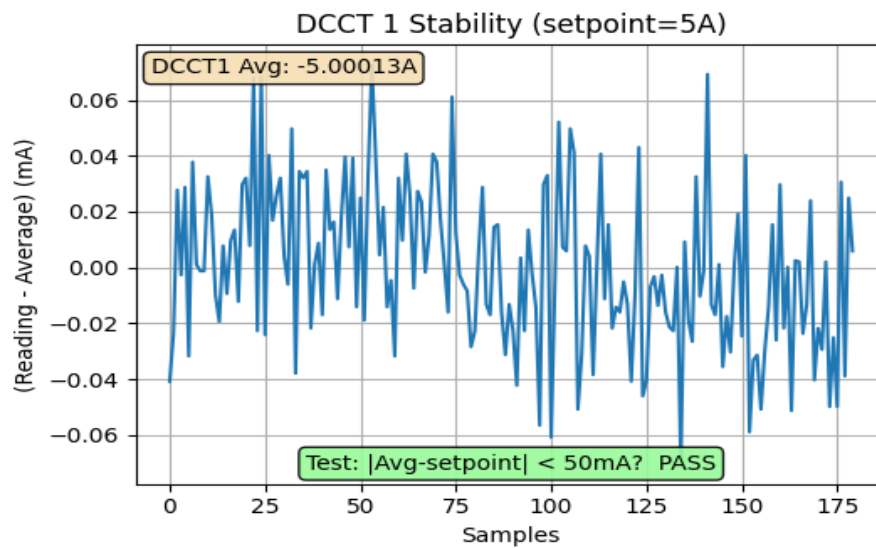
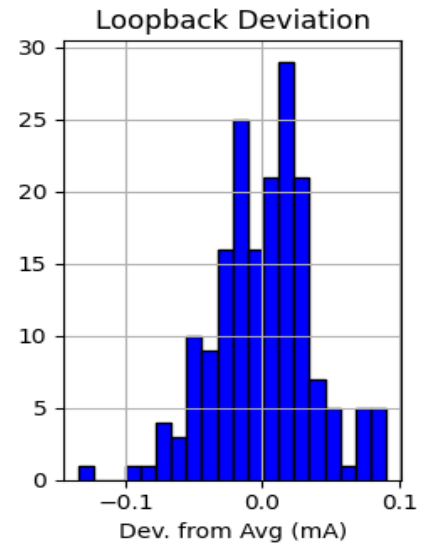
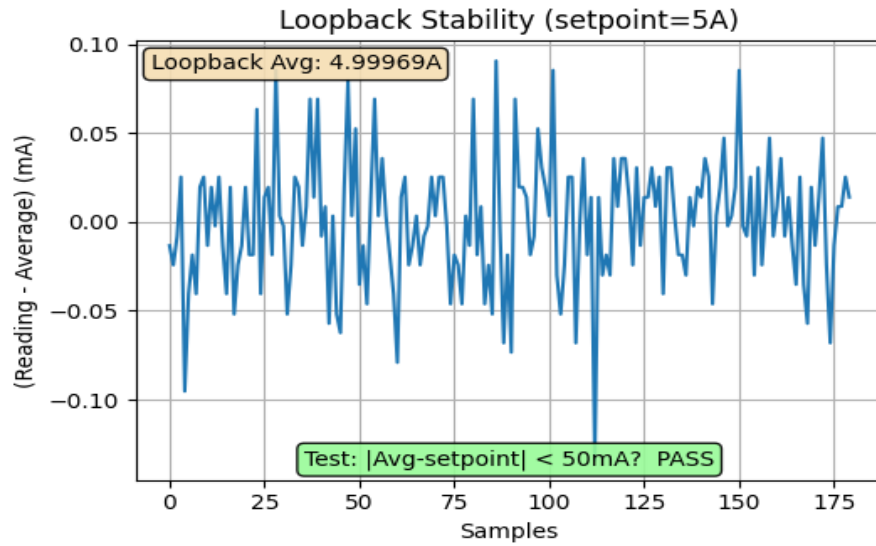
PSC Functional Test Results	
Power Supply Controller Configuration	
Description	PSC-4CH-HSF-EIC
Serial Number	0001
Number of Channels	4
Resolution	HS (20bit)
Bandwidth	F
Polarity	Bipolar



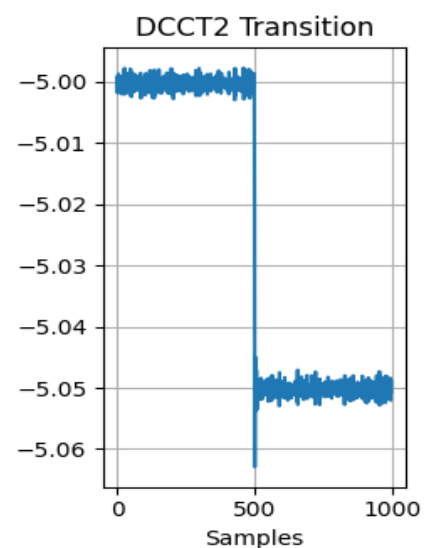
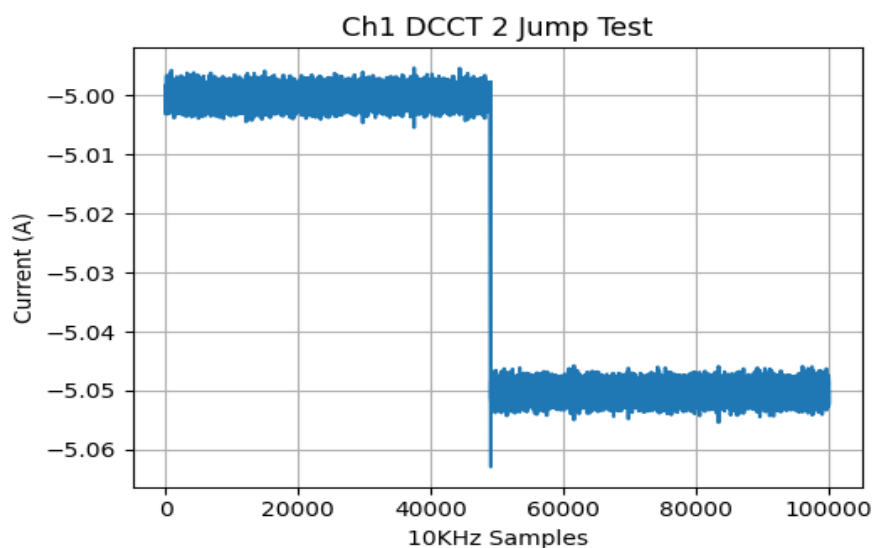
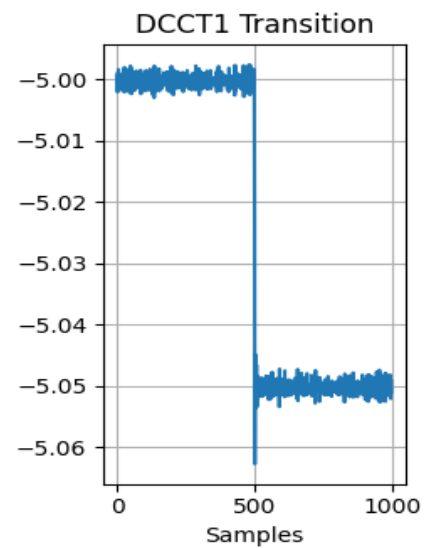
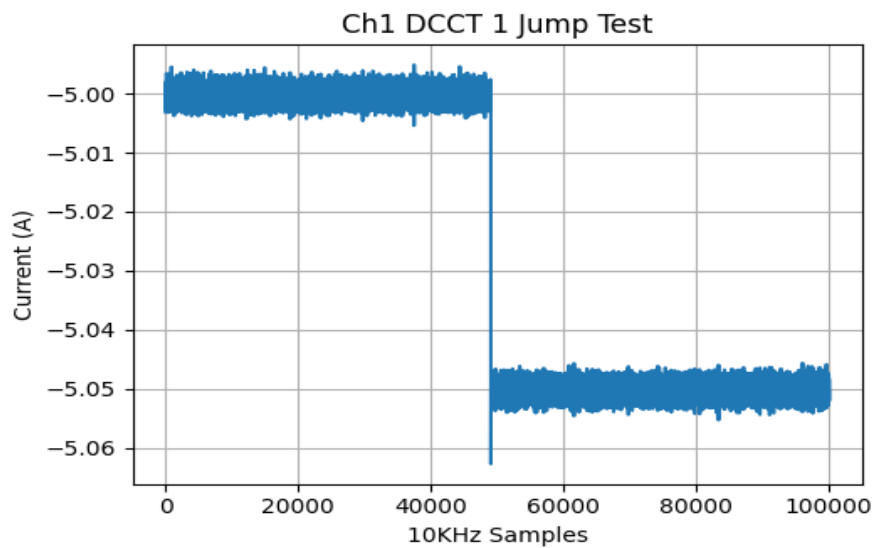
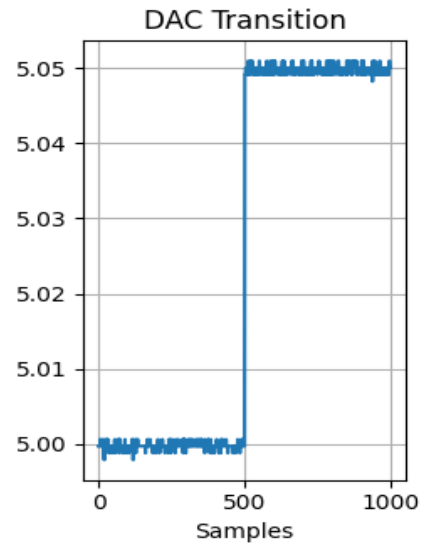
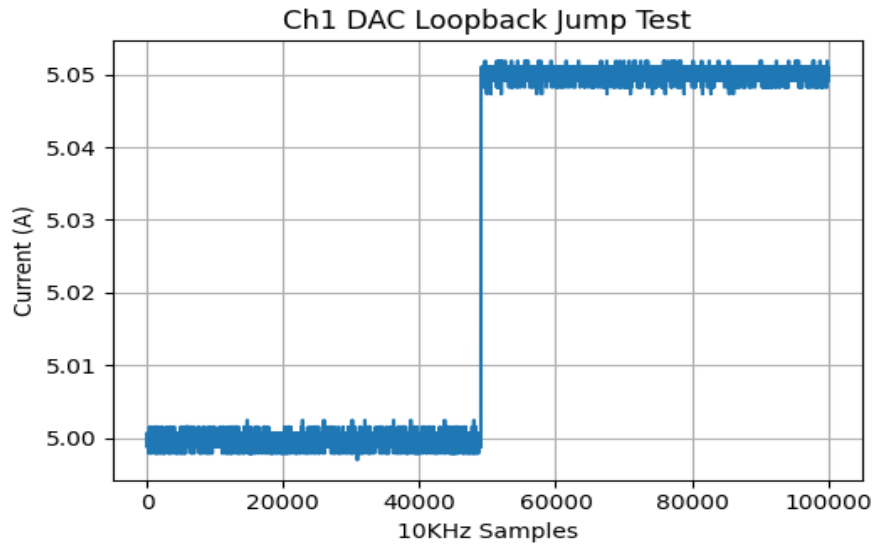
Channel 1

ATE Fault Tests for Channel 1	
Fault #1 Generated and Cleared	PASS
Fault #2 Generated and Cleared	PASS
Fault SPARE Generated and Cleared	PASS
Fault DCCT Generated and Cleared	PASS

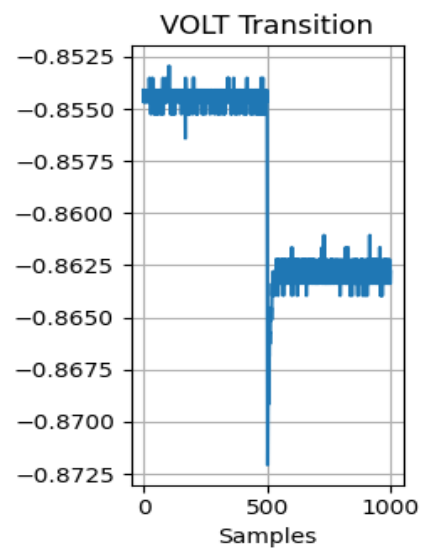
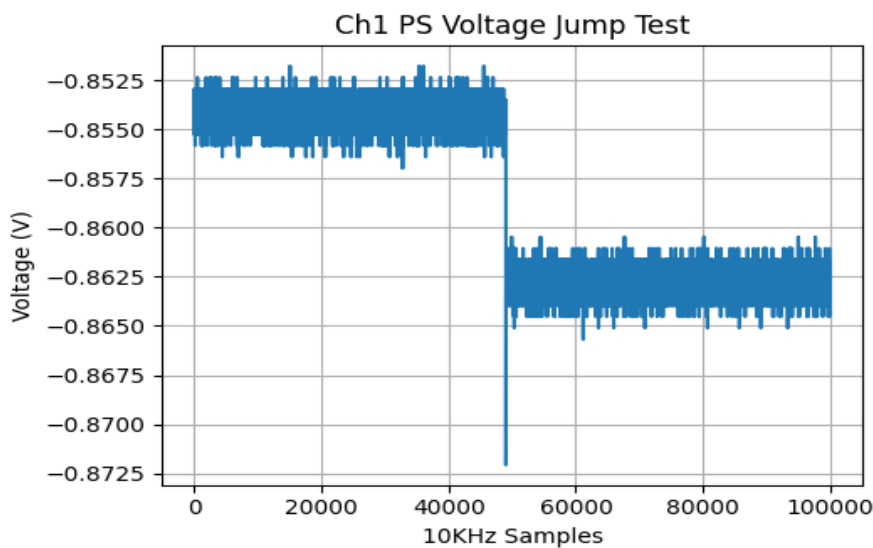
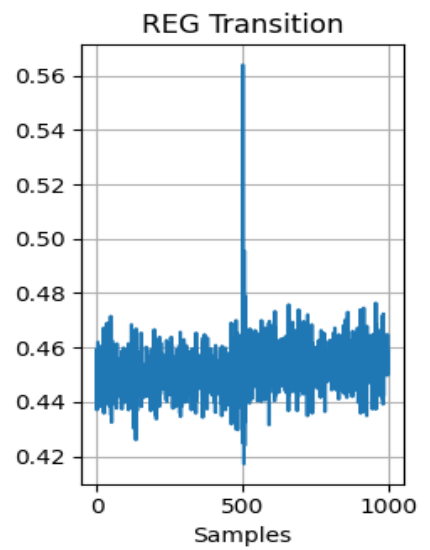
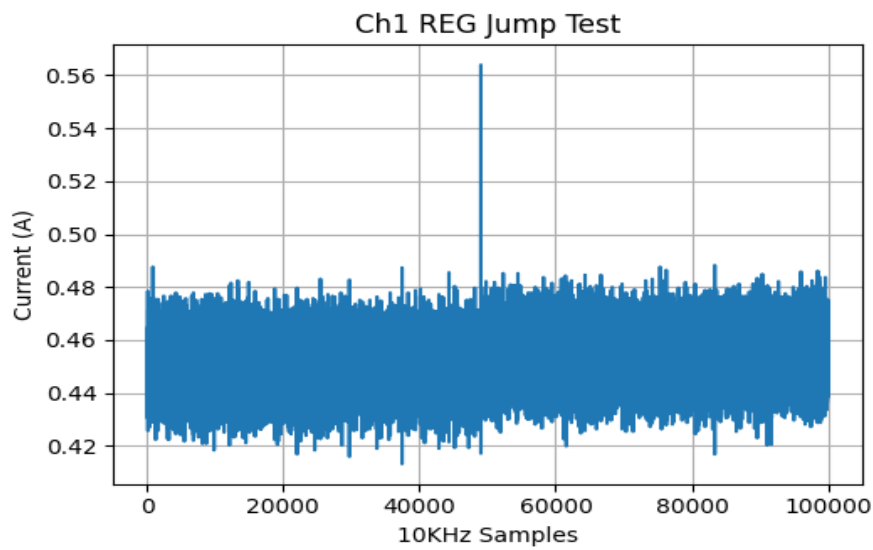
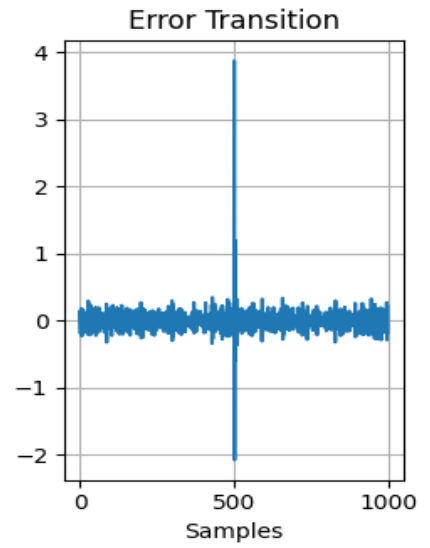
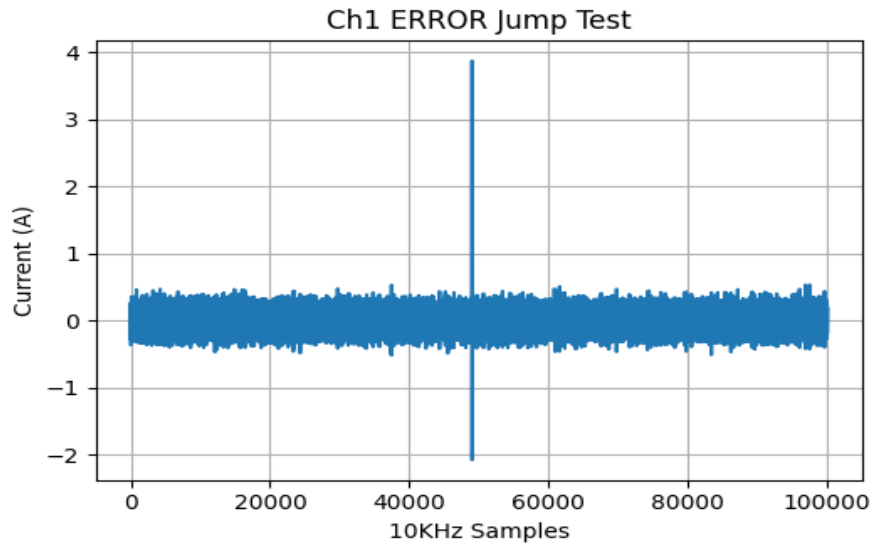
Power Supply Regulation for Channel 1:



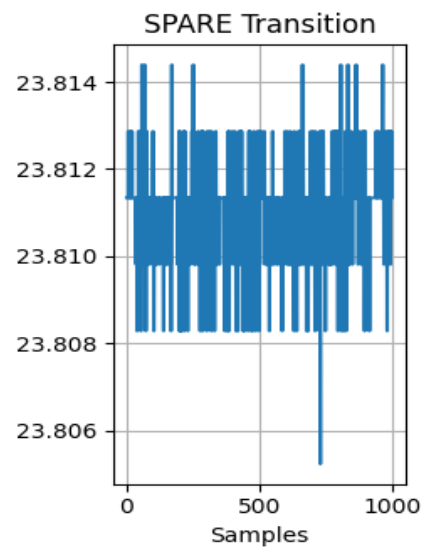
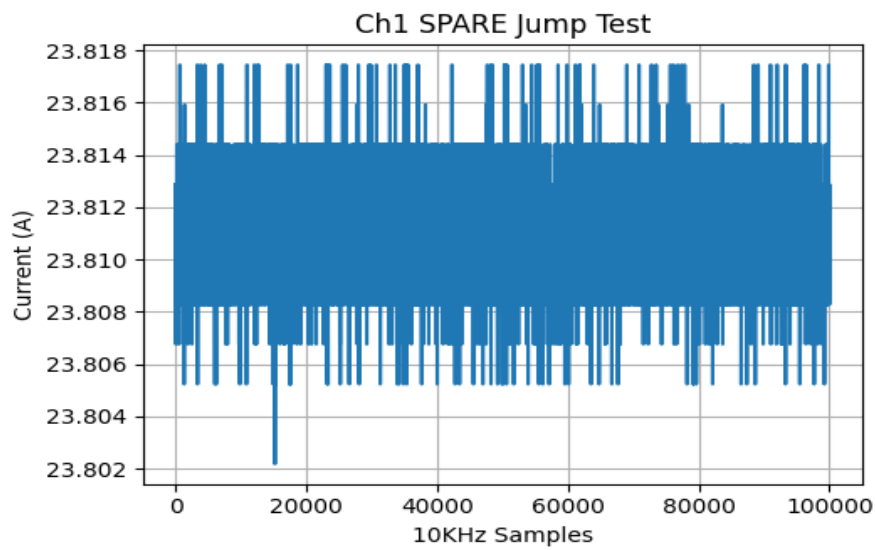
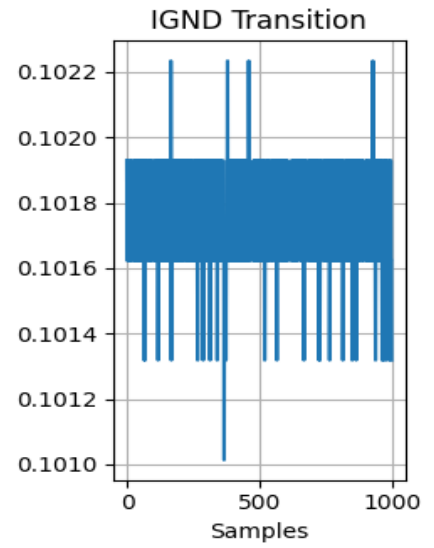
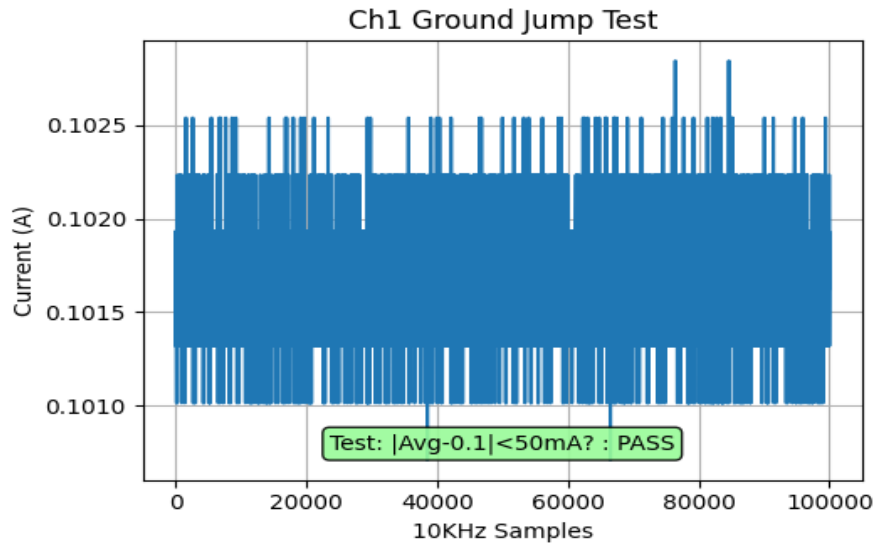
Jump Test Results: CH1



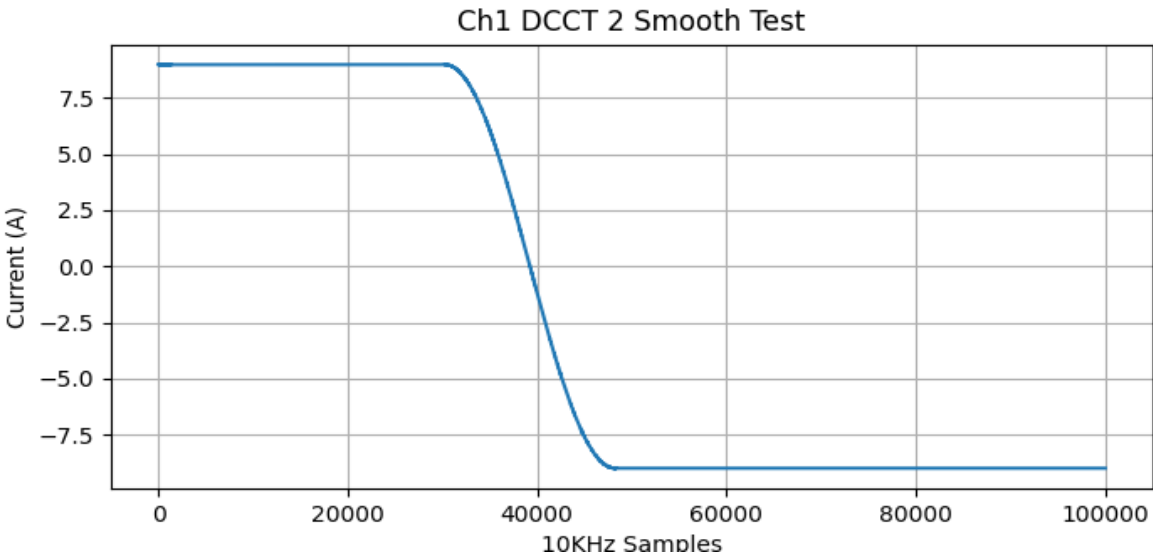
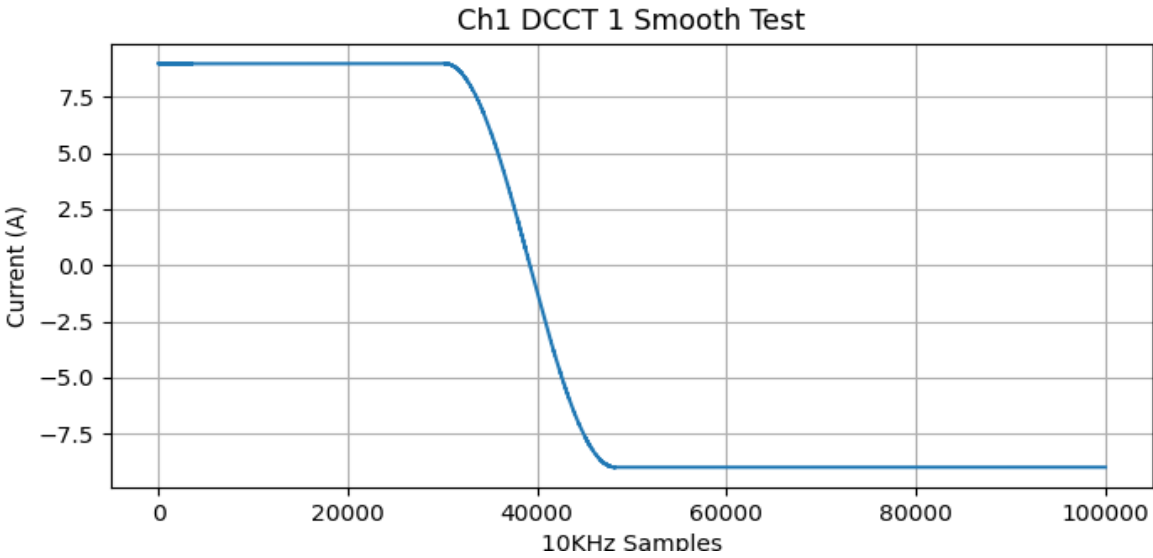
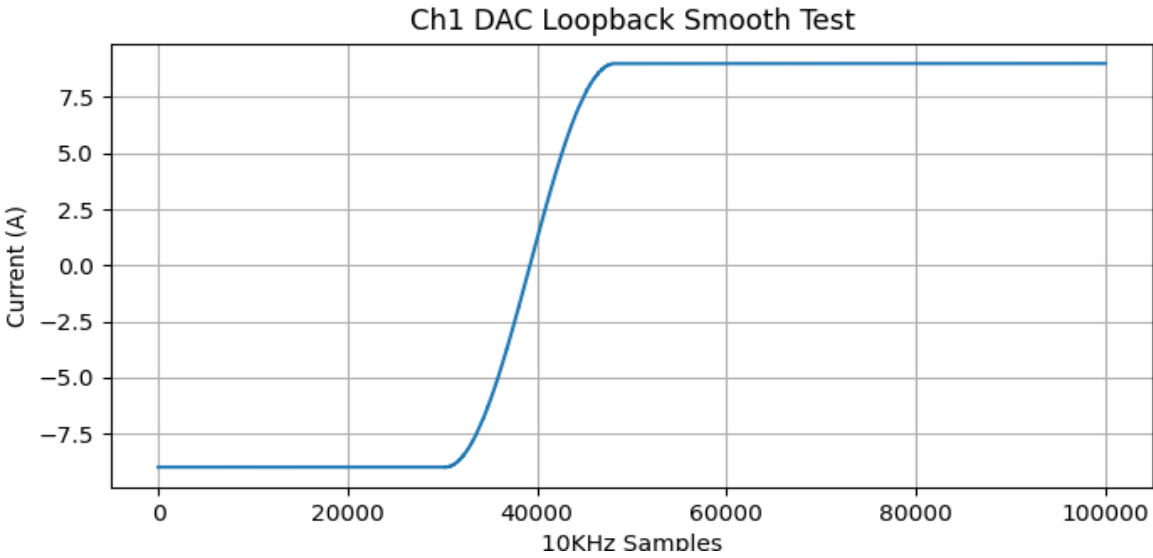
Jump Test Results: CH1



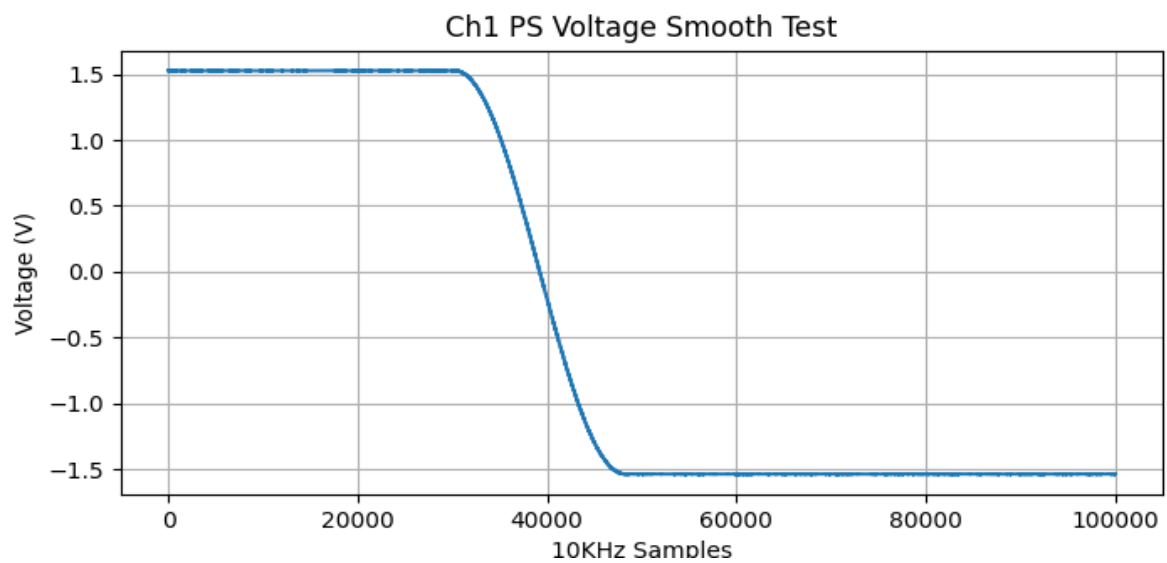
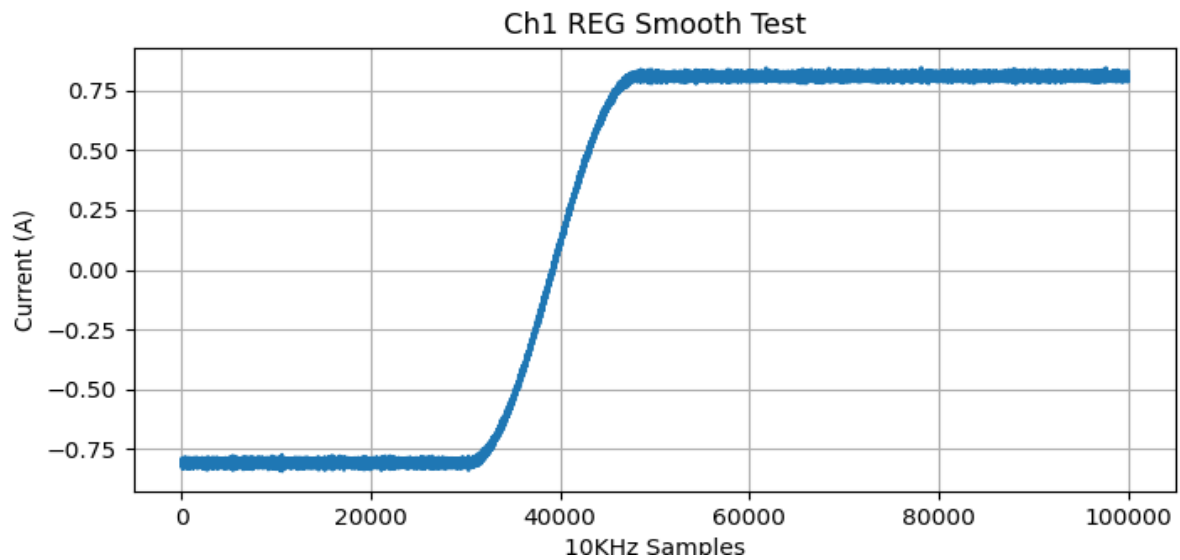
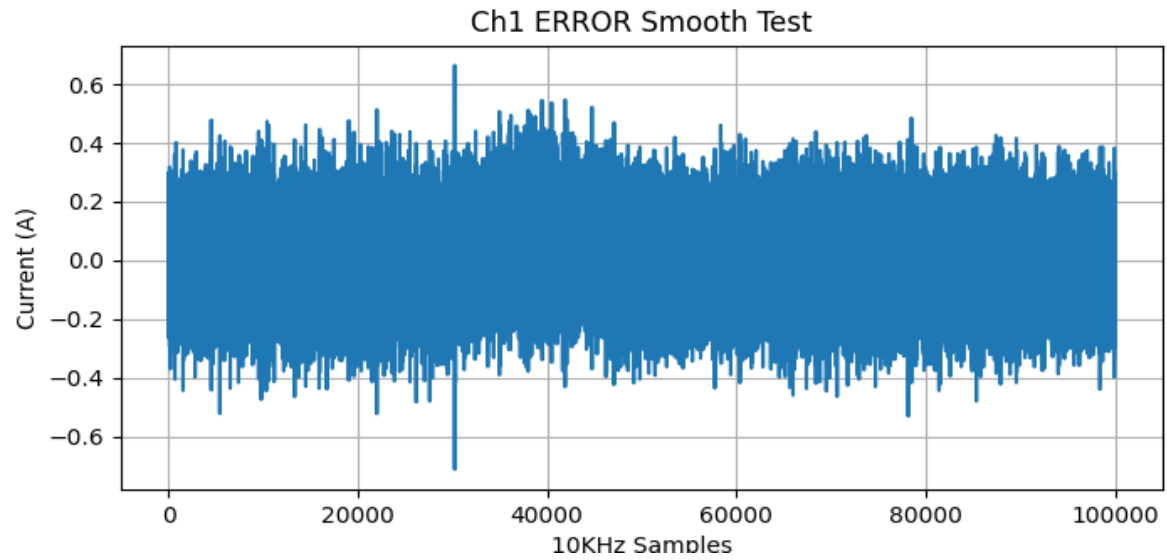
Jump Test Results: CH1



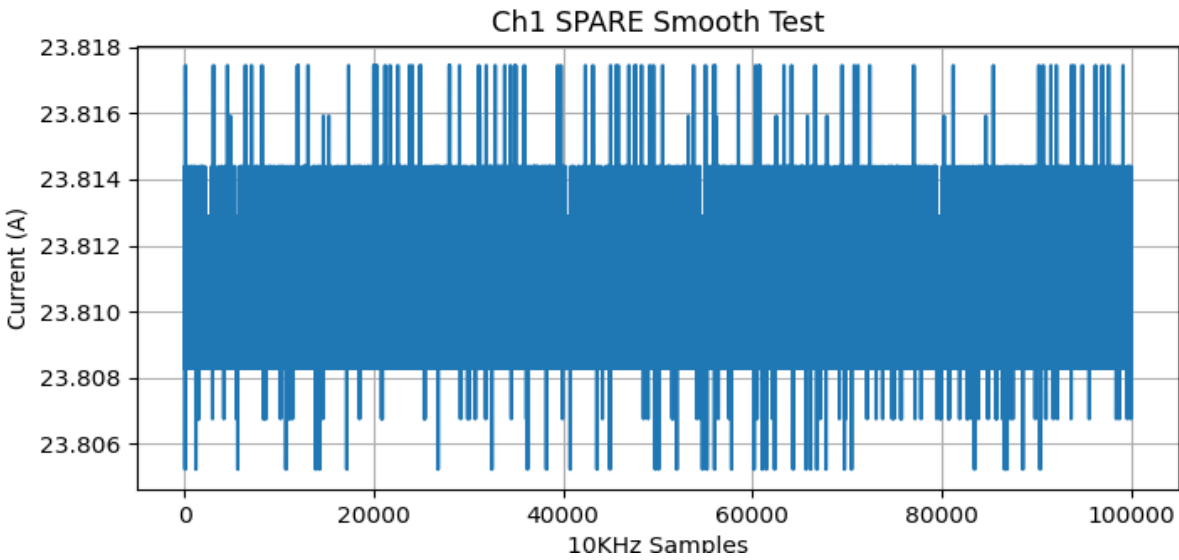
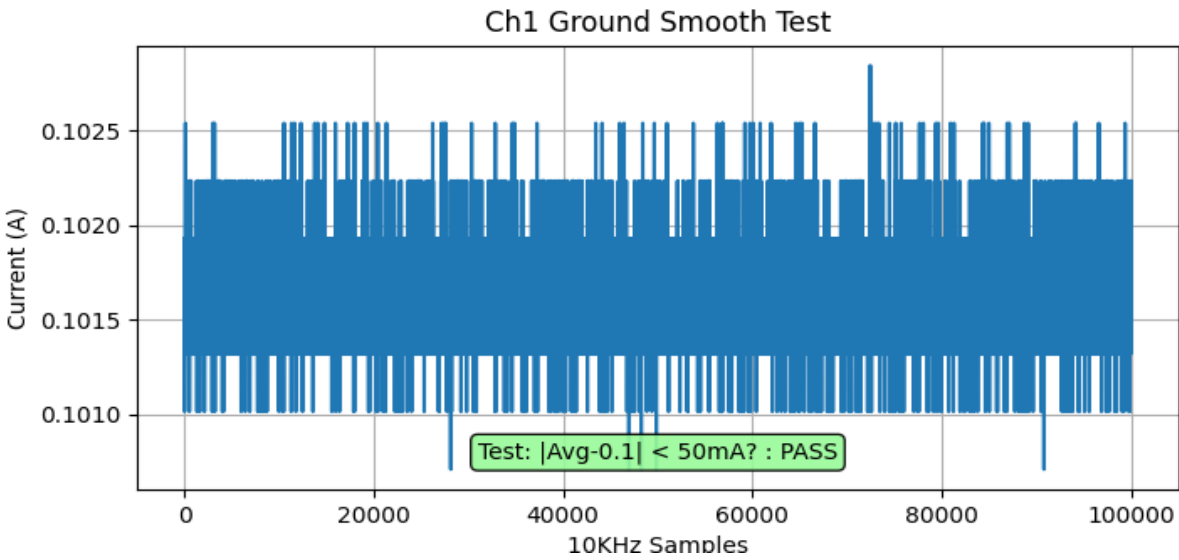
Smooth Test Results: CH1



Smooth Test Results: CH1



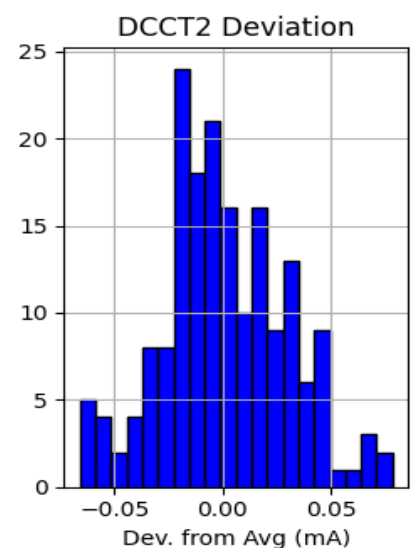
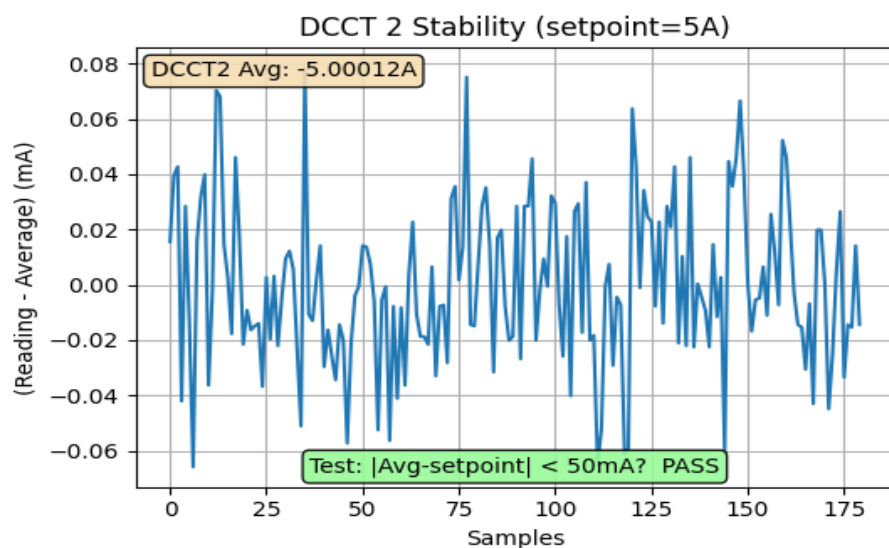
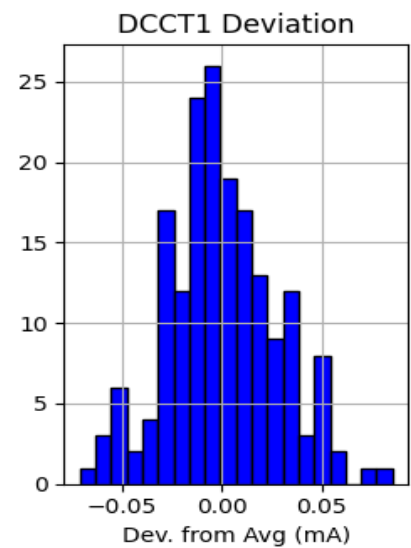
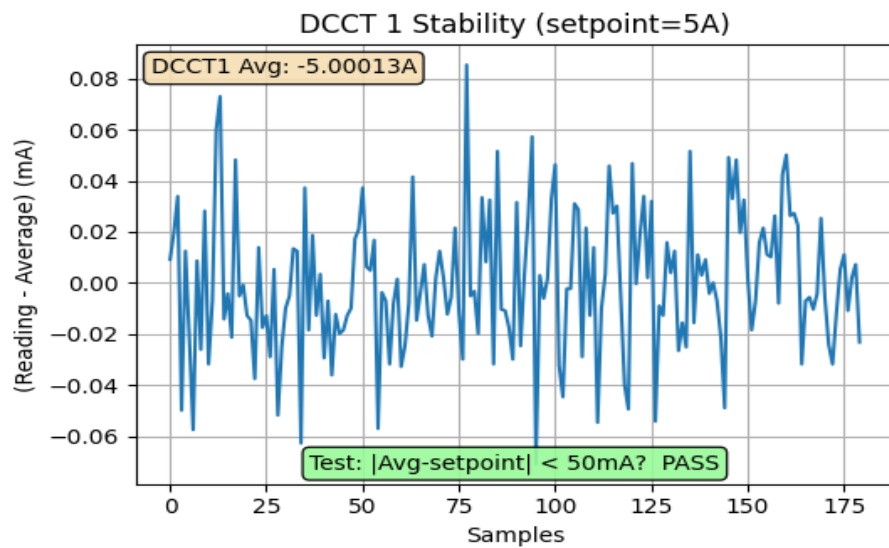
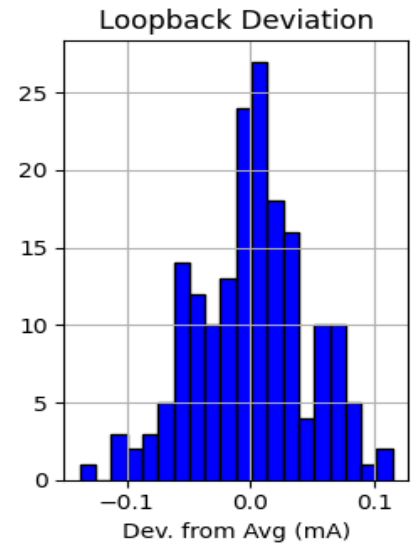
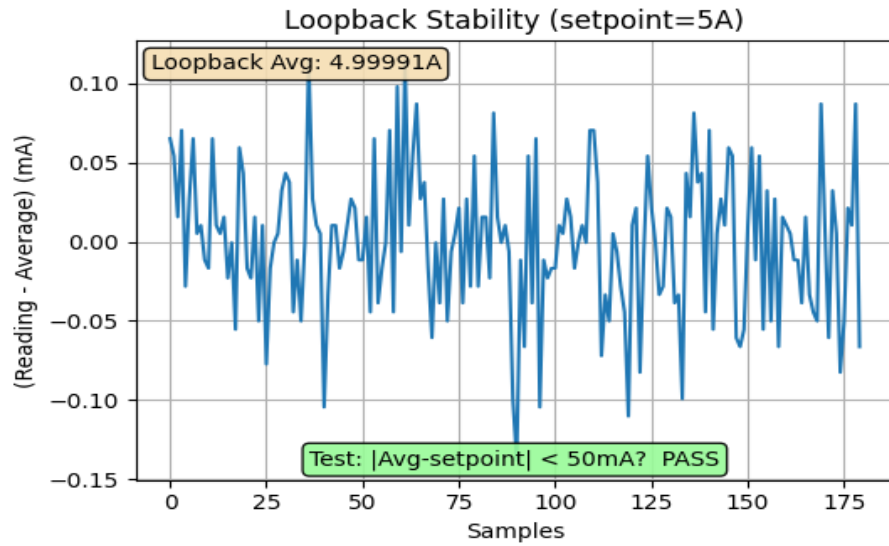
Smooth Test Results: CH1



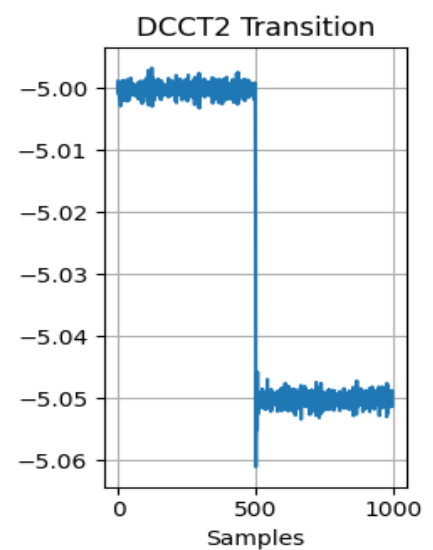
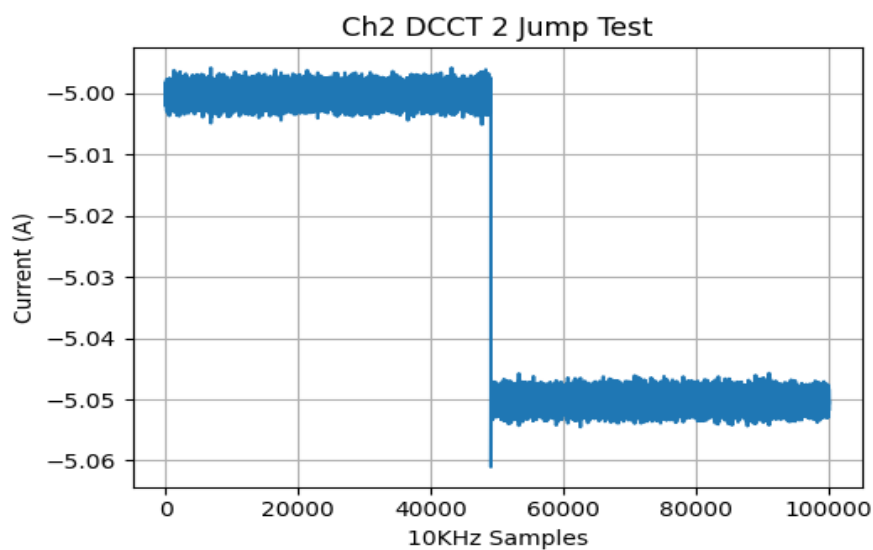
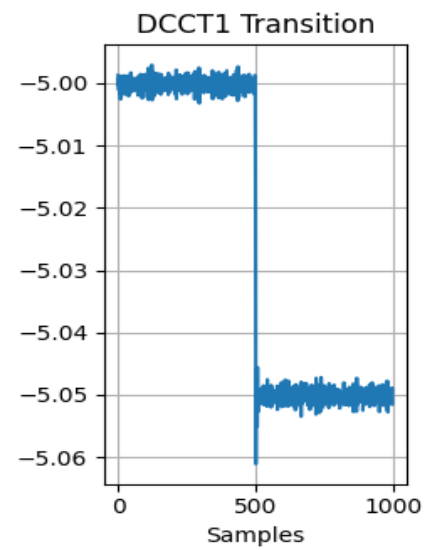
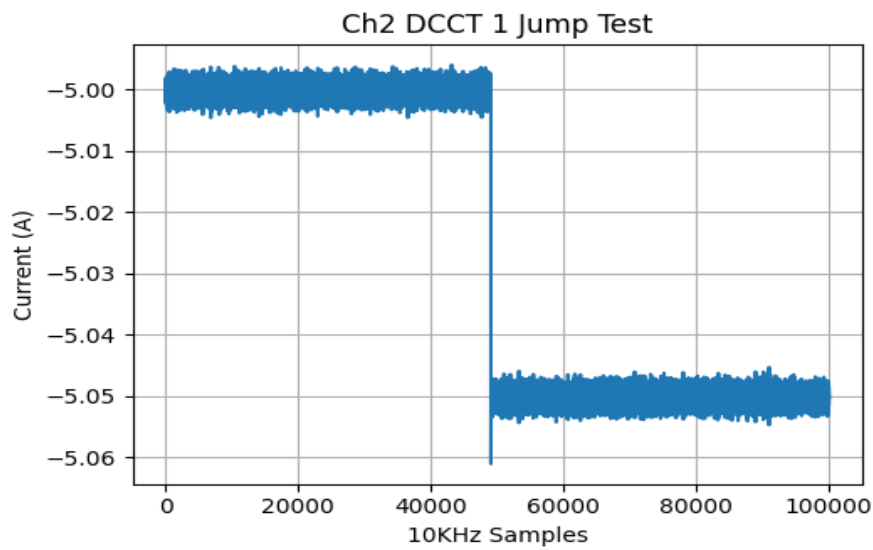
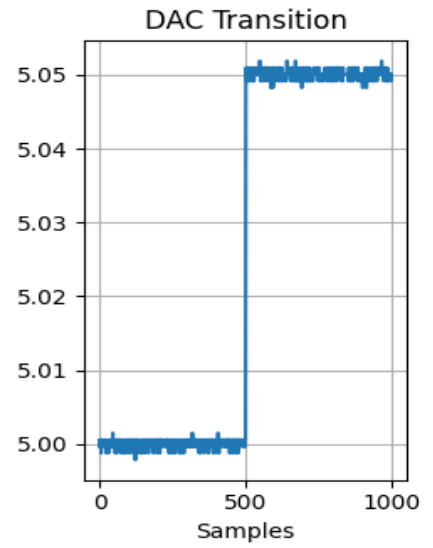
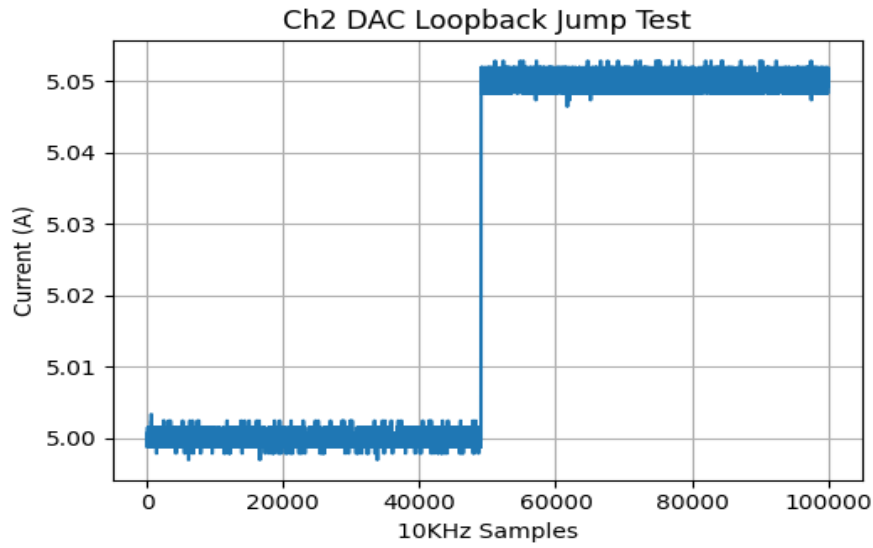
Channel 2

ATE Fault Tests for Channel 2	
Fault #1 Generated and Cleared	PASS
Fault #2 Generated and Cleared	PASS
Fault SPARE Generated and Cleared	PASS
Fault DCCT Generated and Cleared	PASS

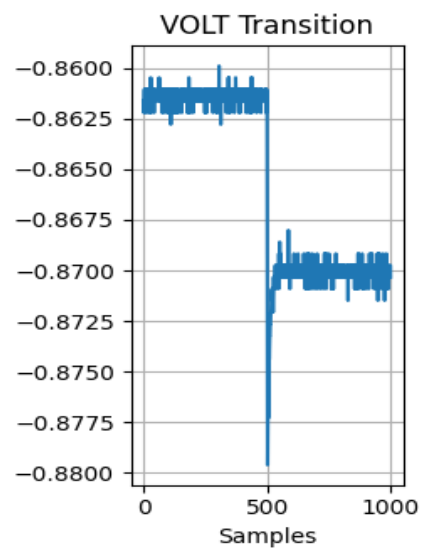
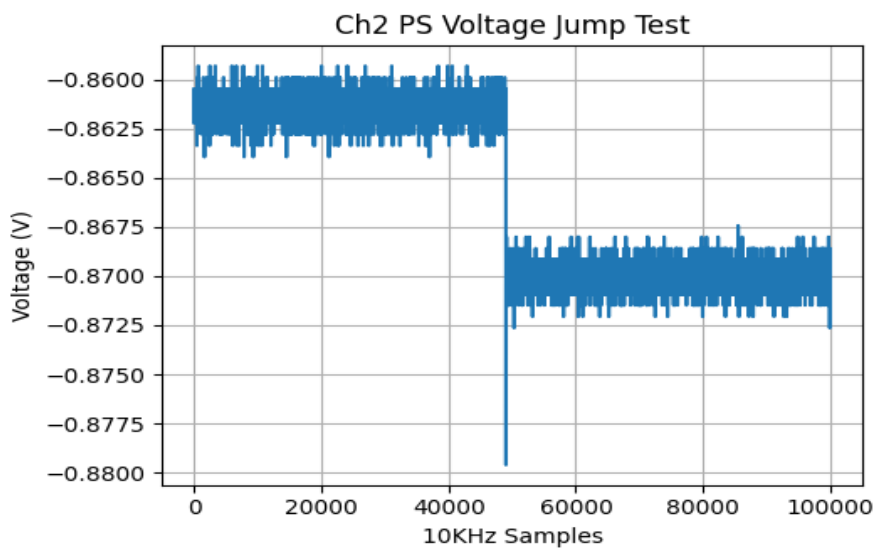
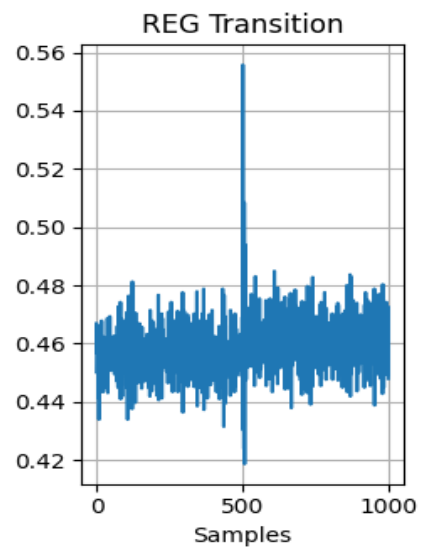
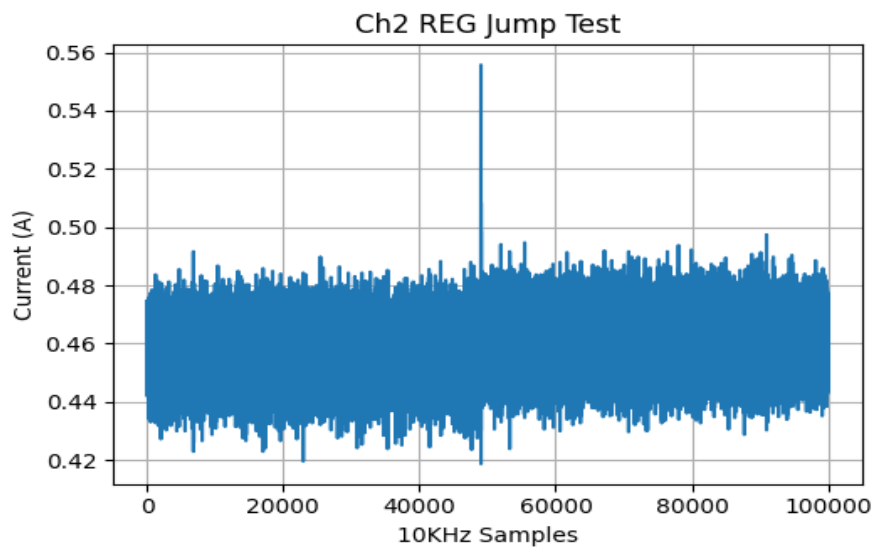
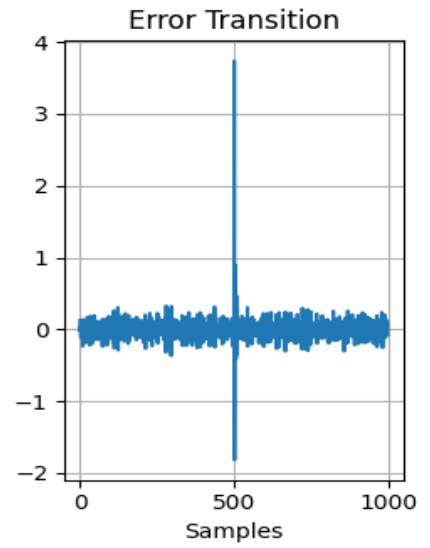
Power Supply Regulation for Channel 2:



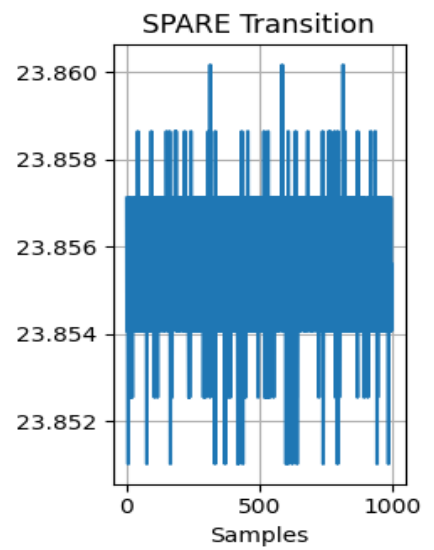
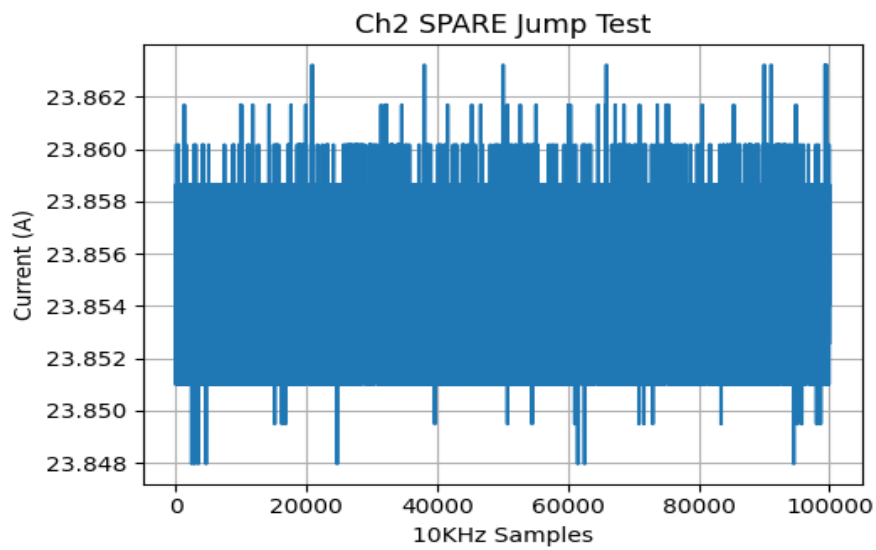
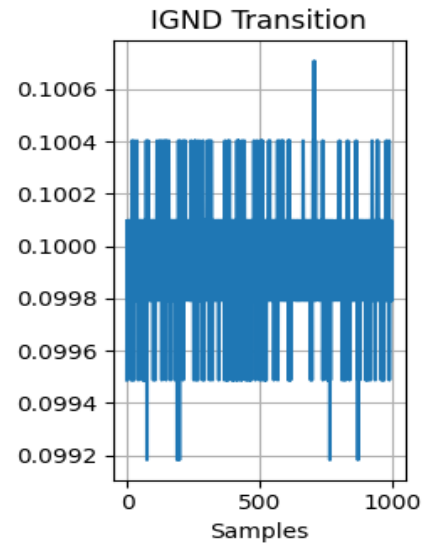
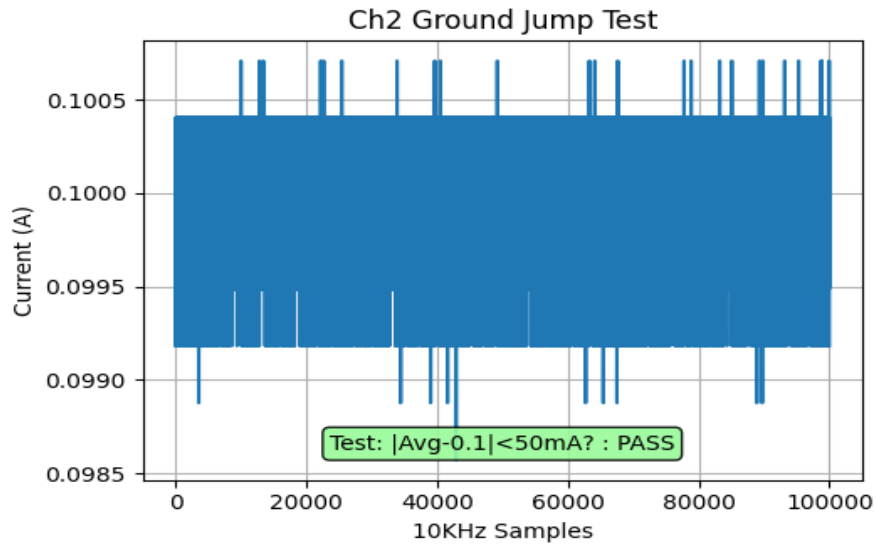
Jump Test Results: CH2



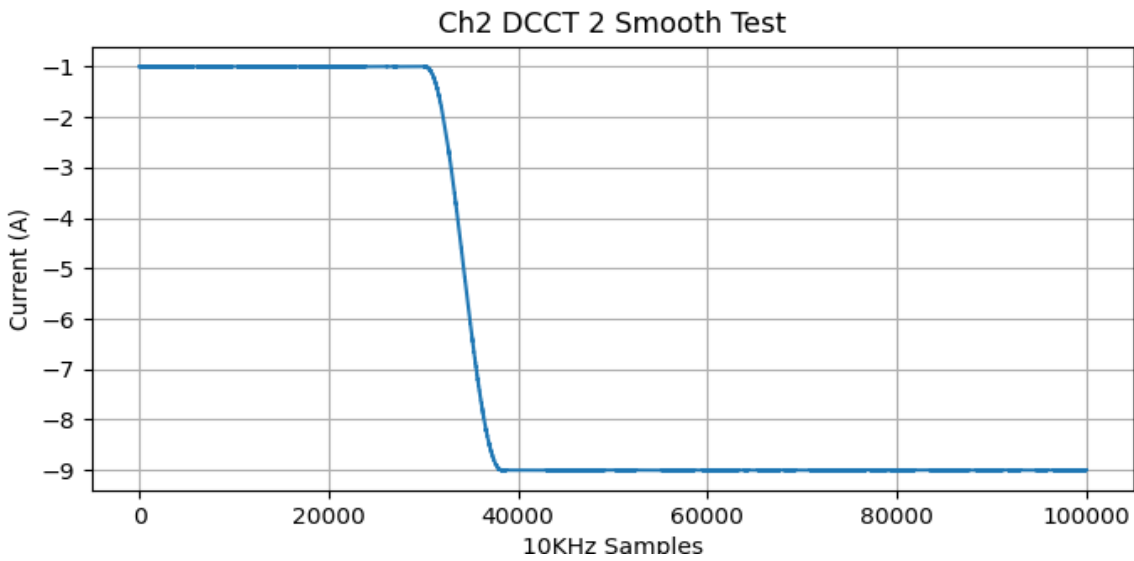
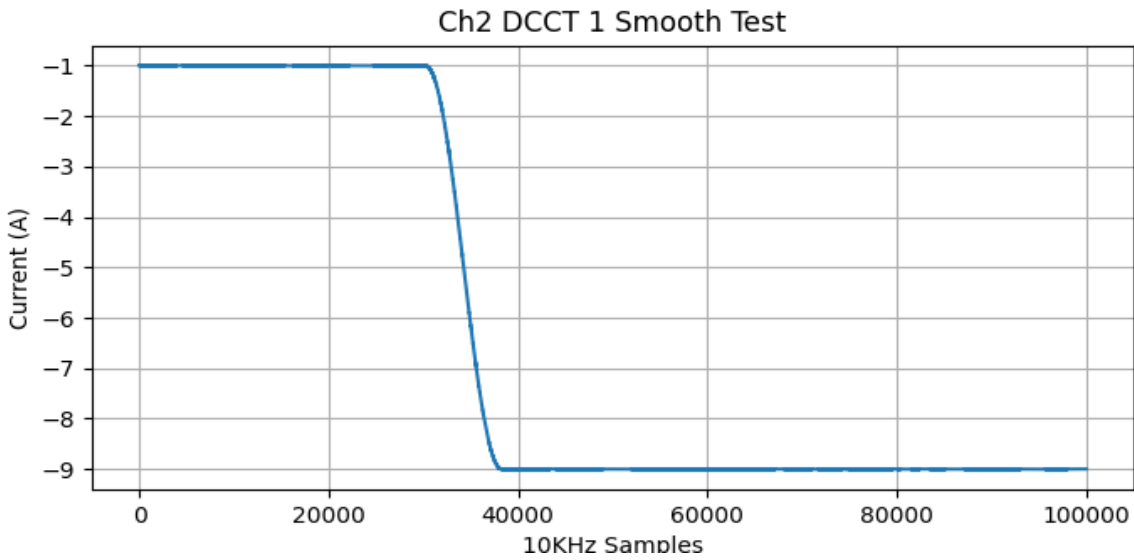
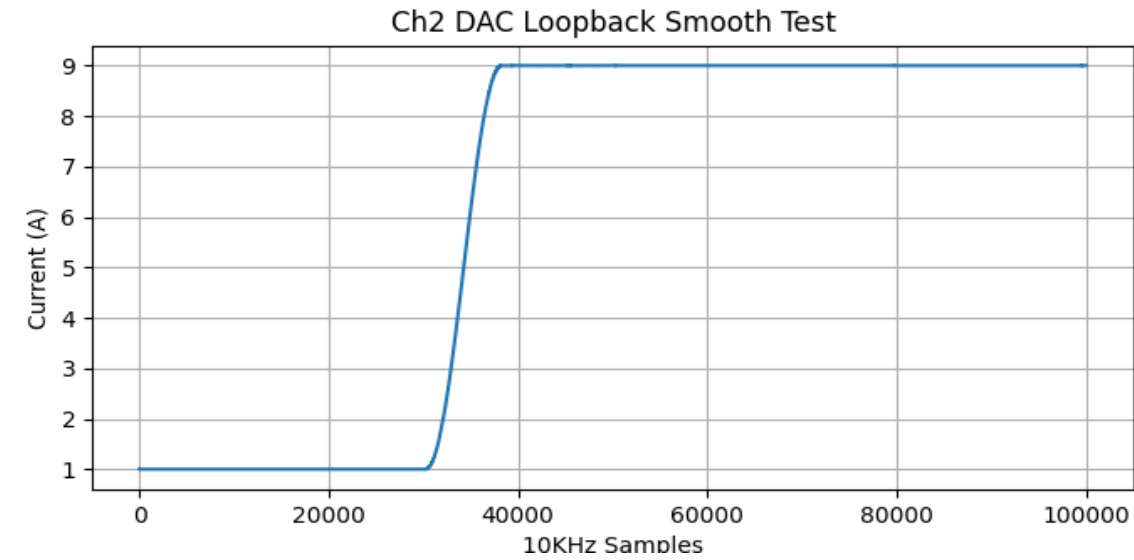
Jump Test Results: CH2



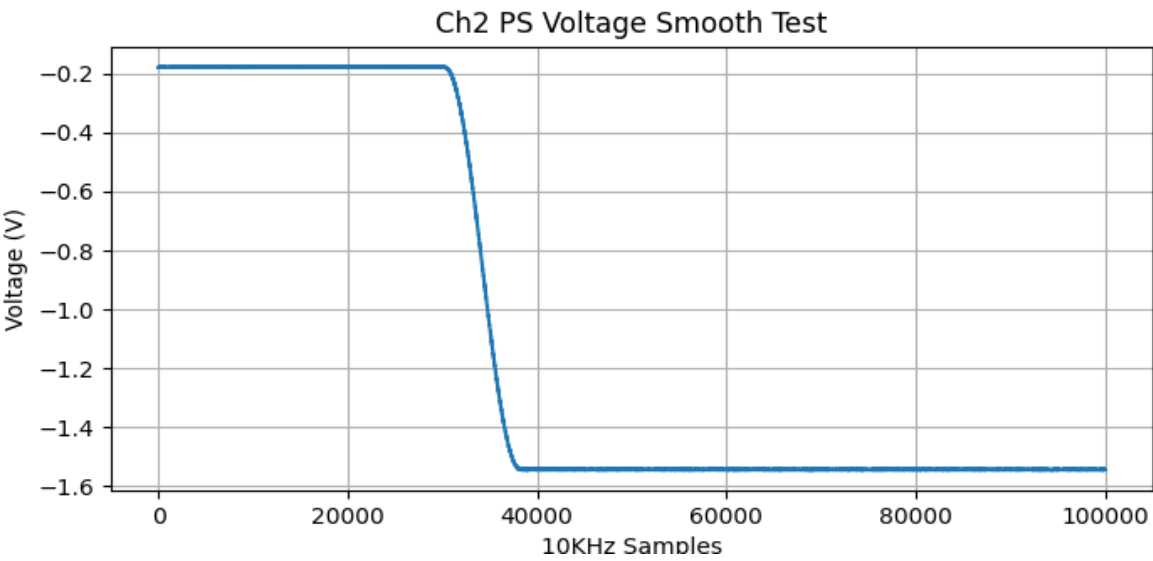
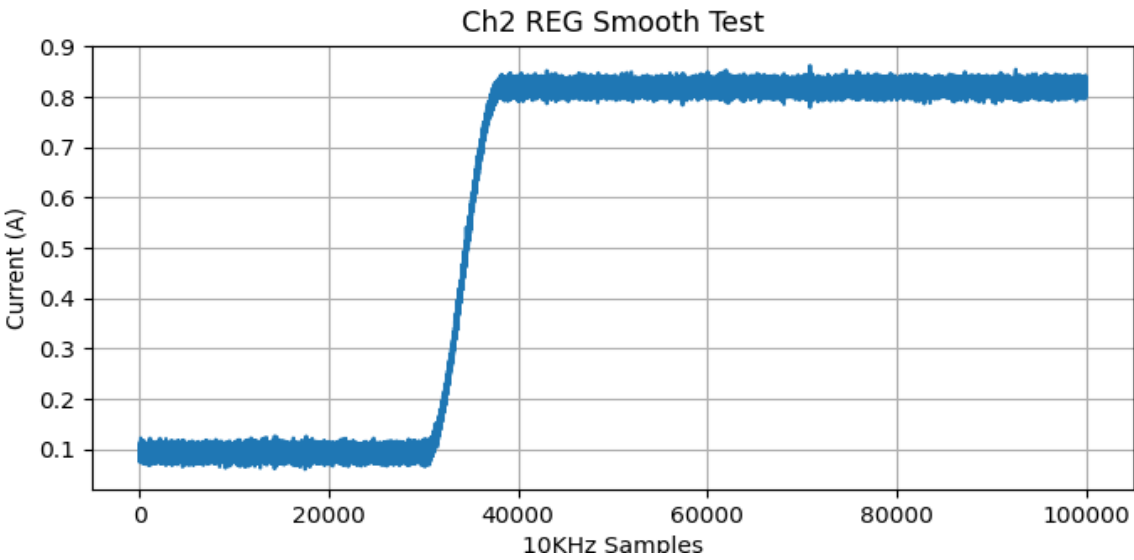
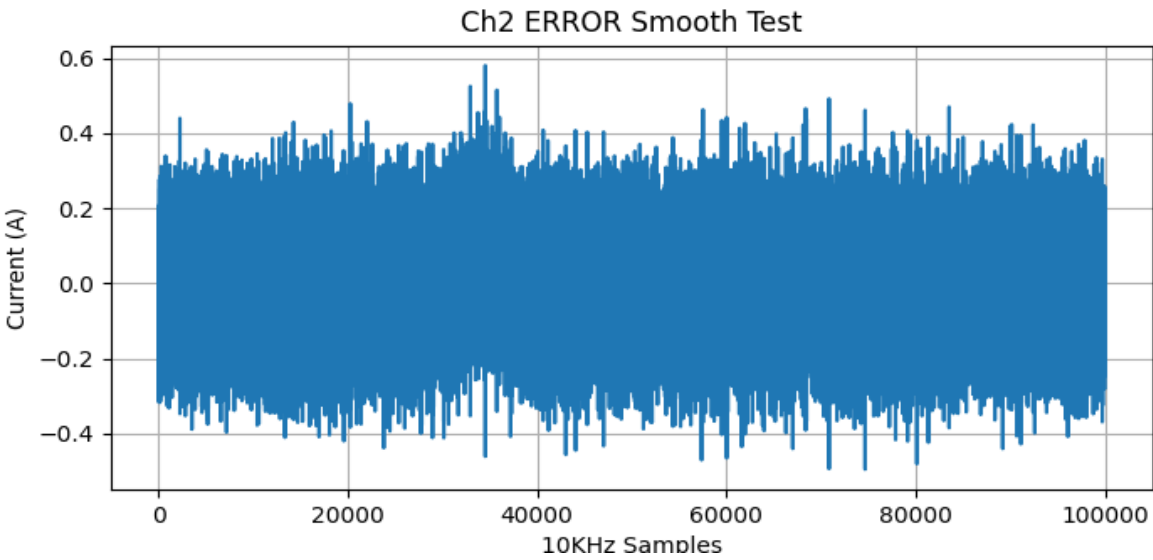
Jump Test Results: CH2



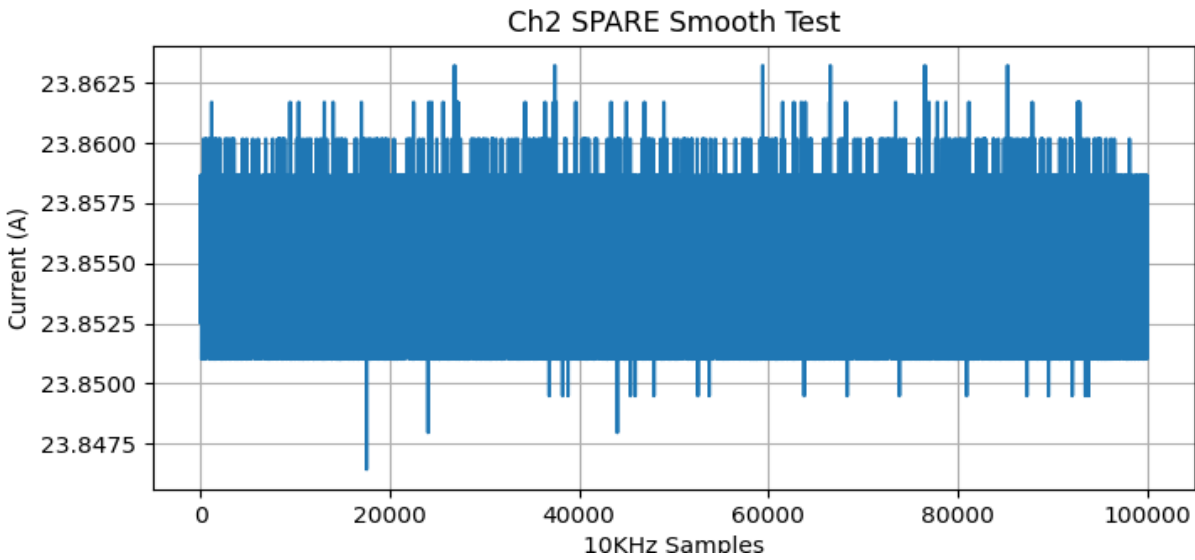
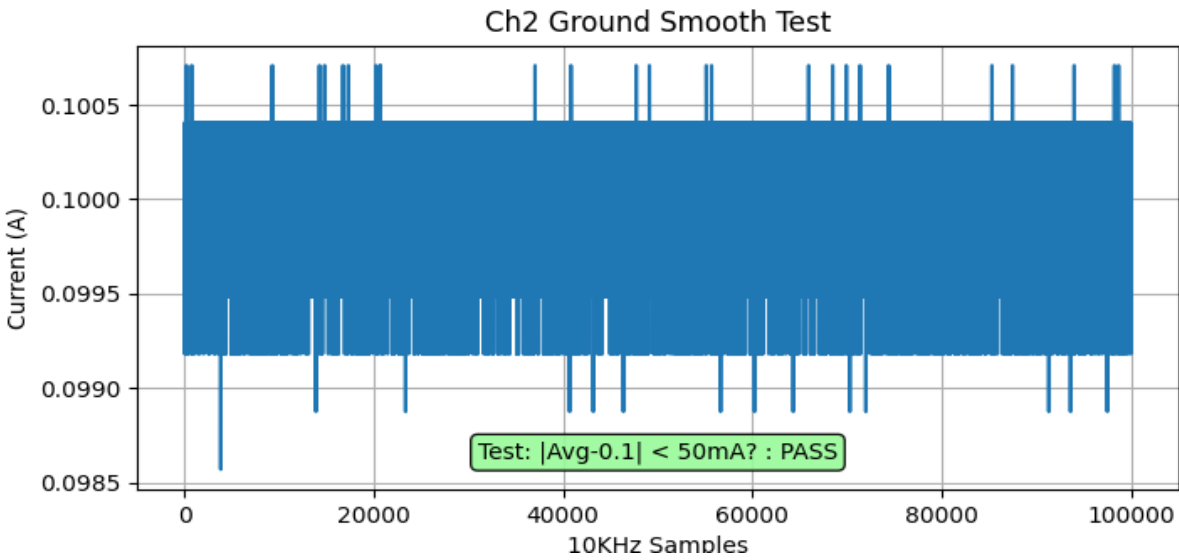
Smooth Test Results: CH2



Smooth Test Results: CH2



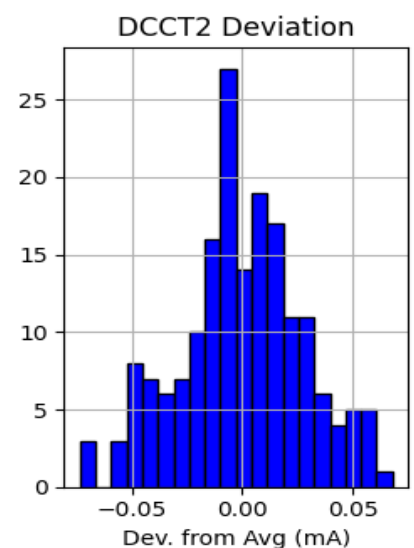
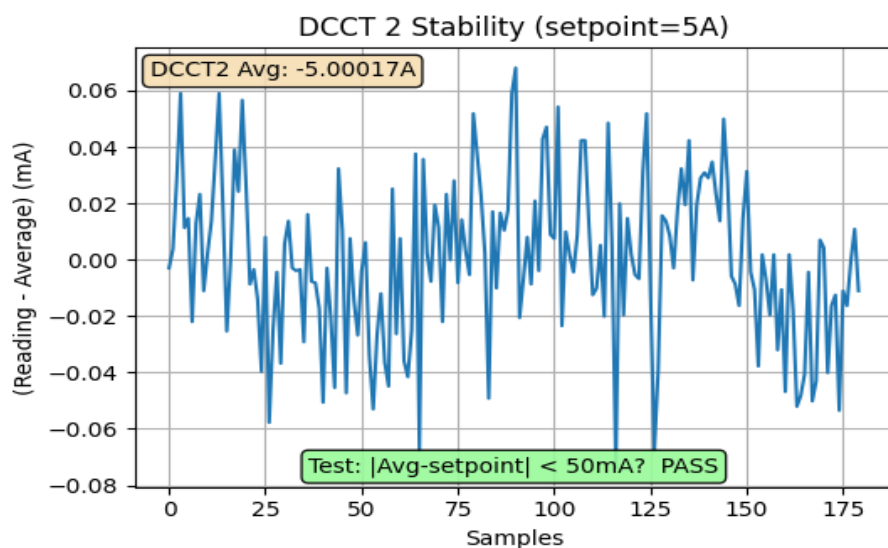
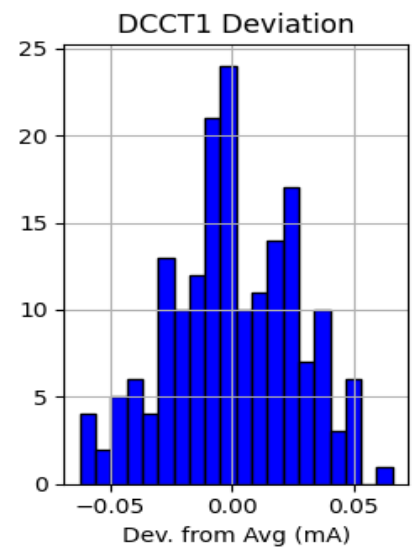
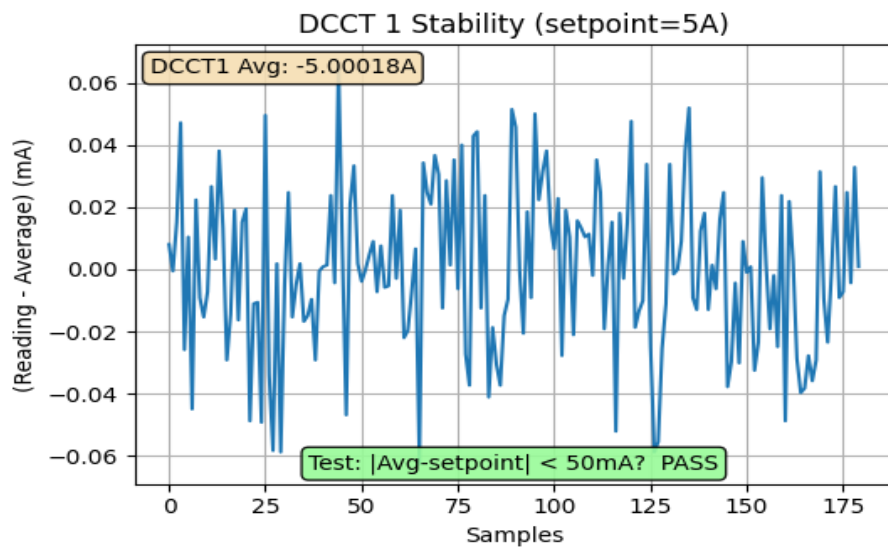
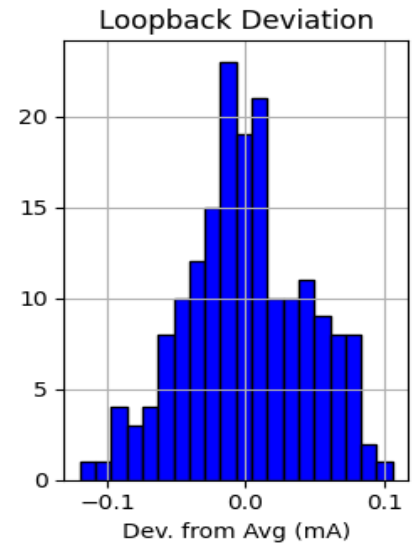
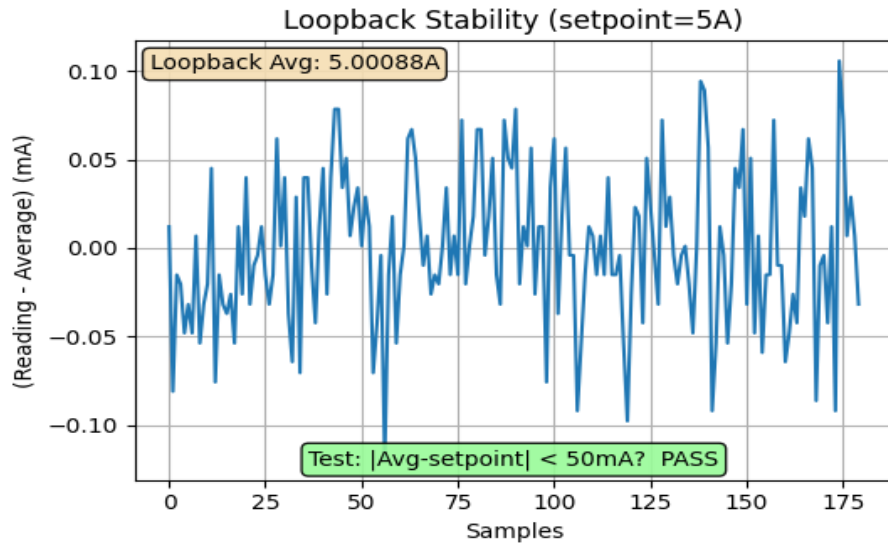
Smooth Test Results: CH2



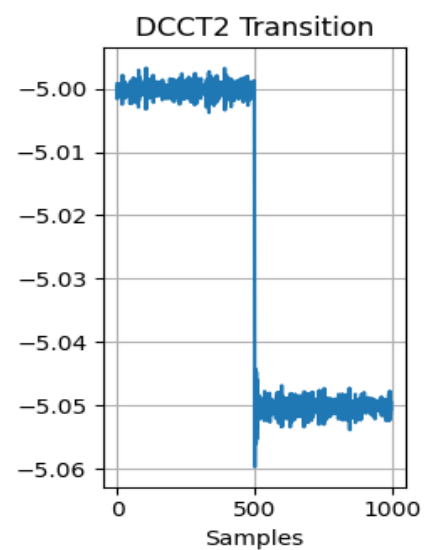
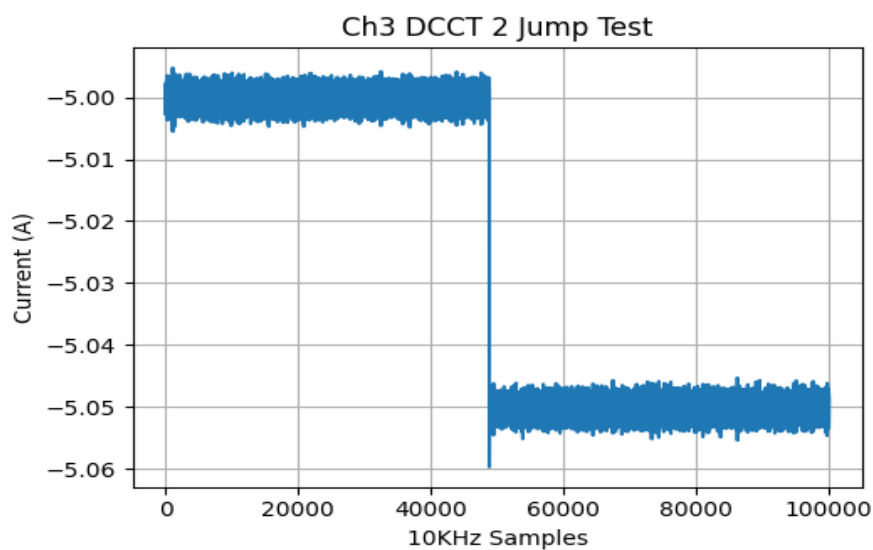
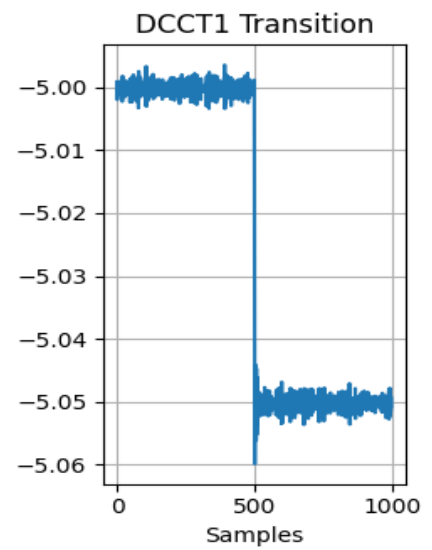
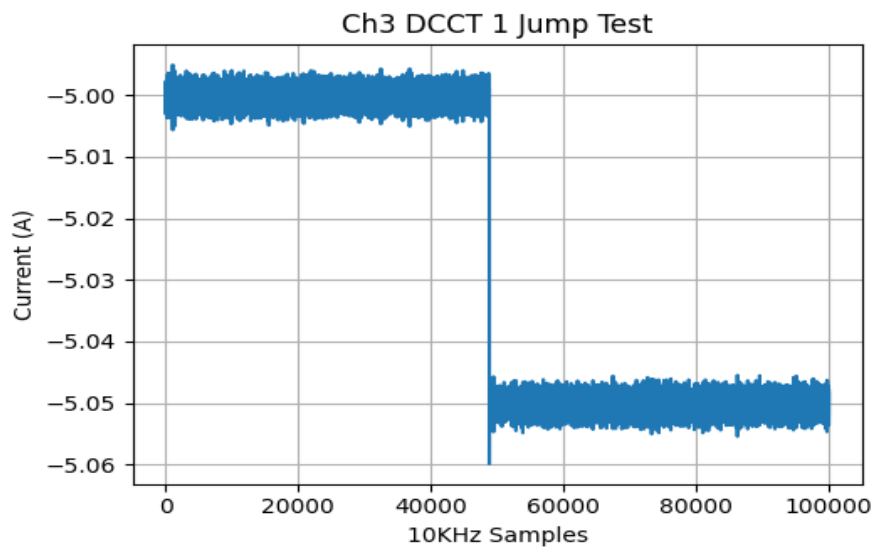
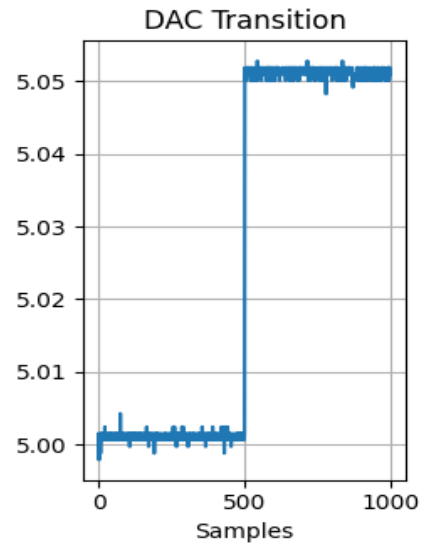
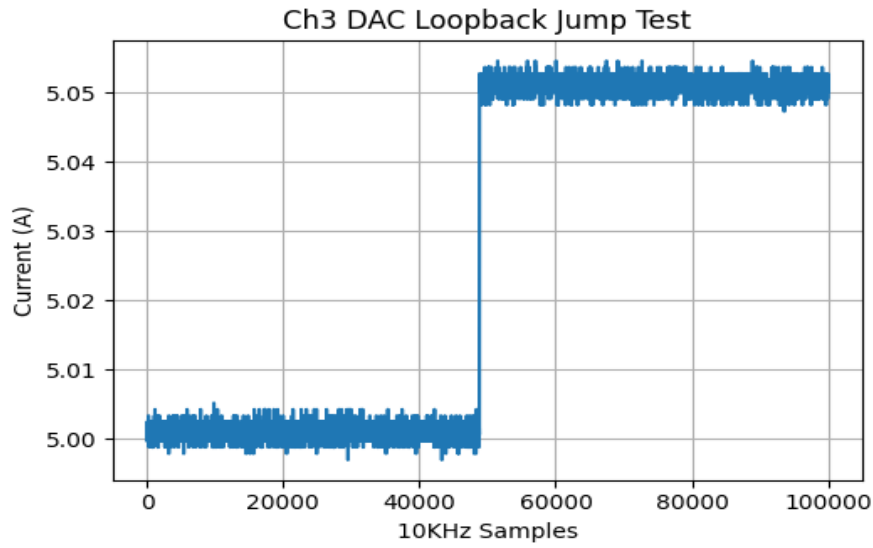
Channel 3

ATE Fault Tests for Channel 3	
Fault #1 Generated and Cleared	PASS
Fault #2 Generated and Cleared	PASS
Fault SPARE Generated and Cleared	PASS
Fault DCCT Generated and Cleared	PASS

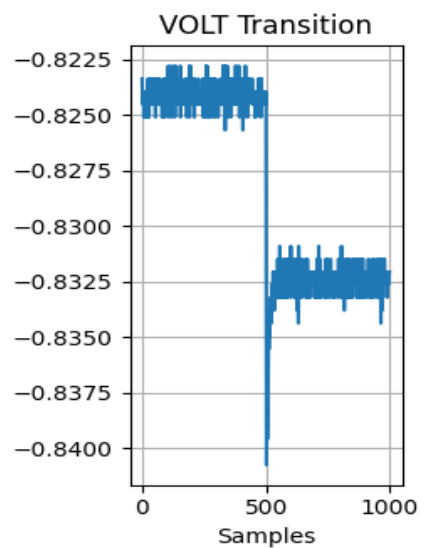
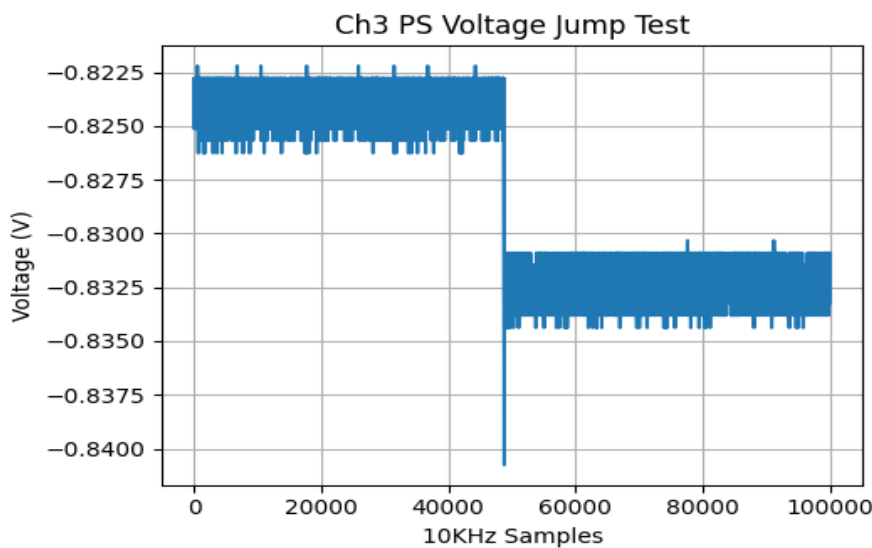
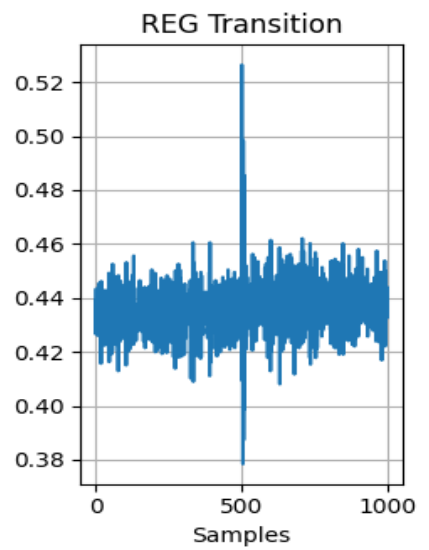
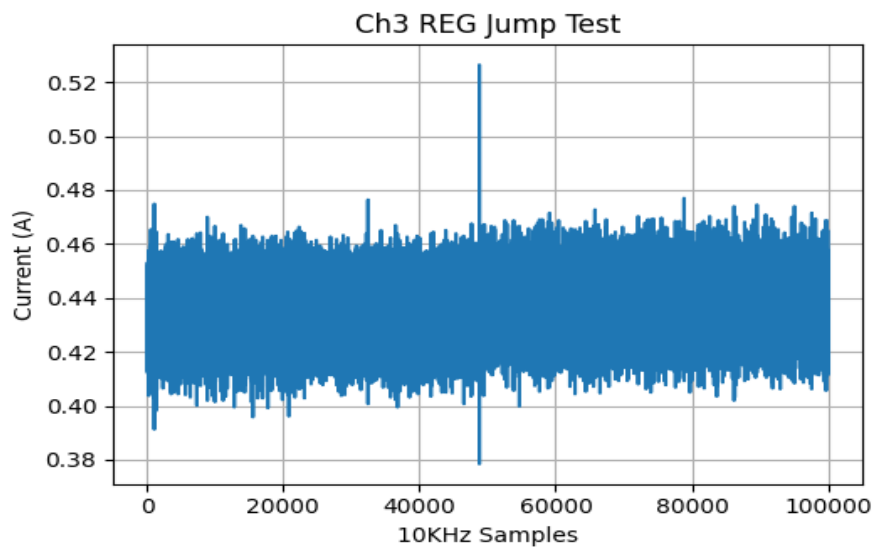
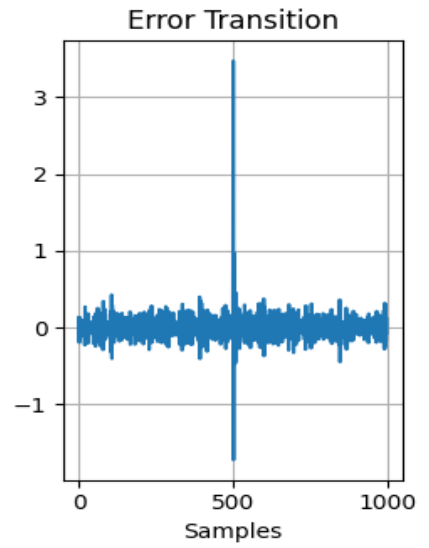
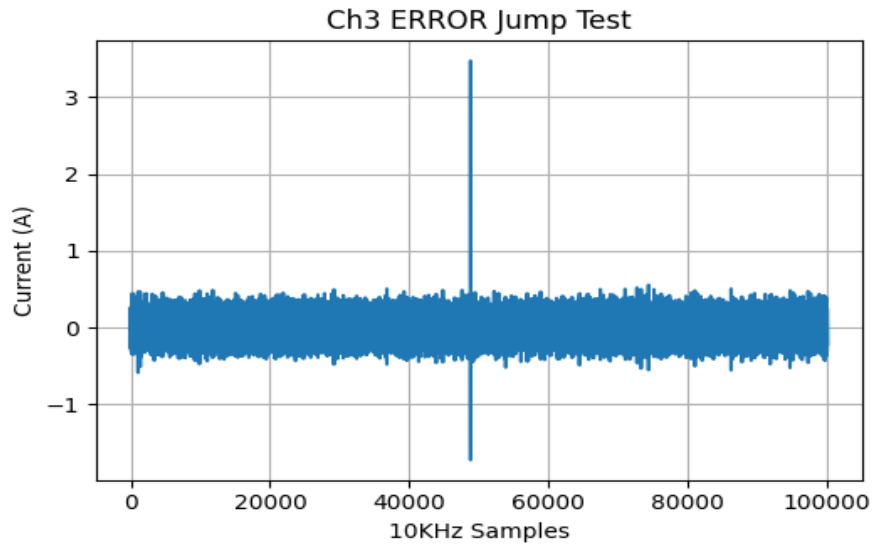
Power Supply Regulation for Channel 3:



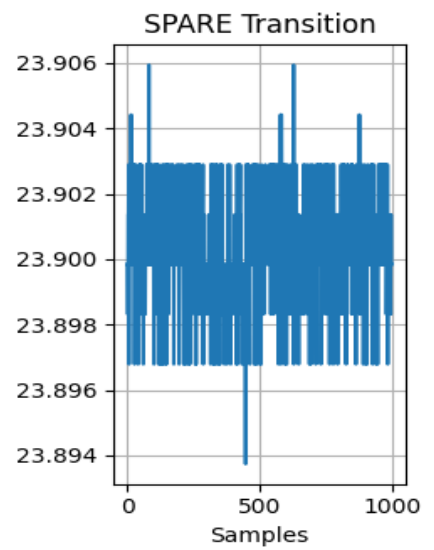
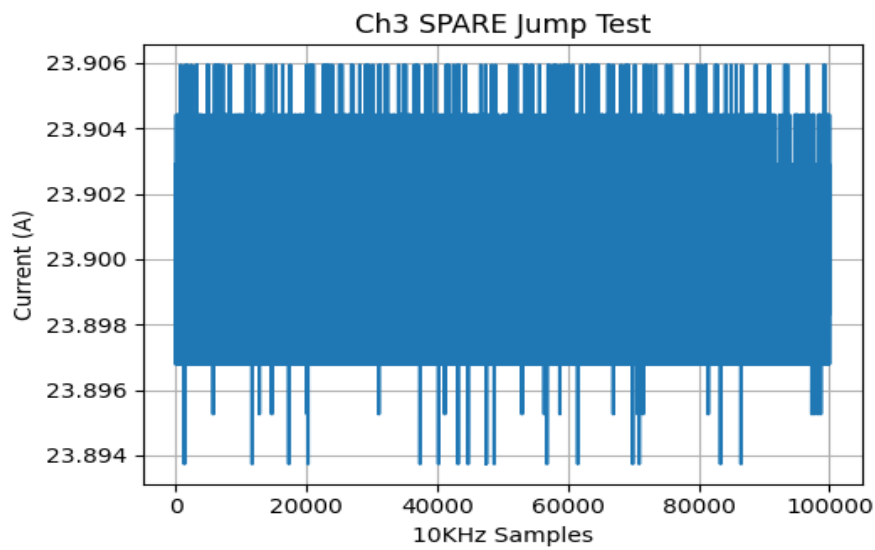
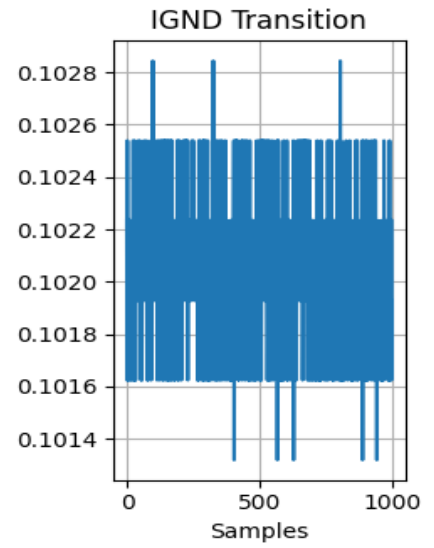
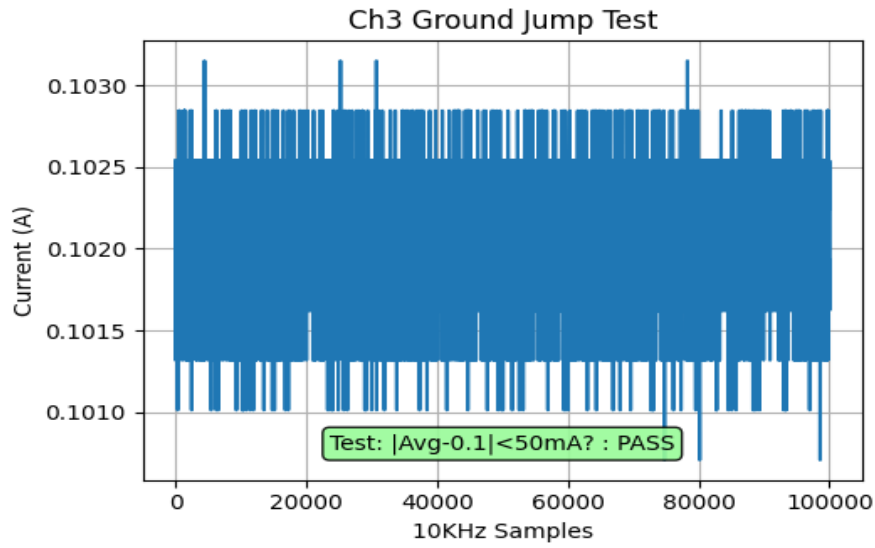
Jump Test Results: CH3



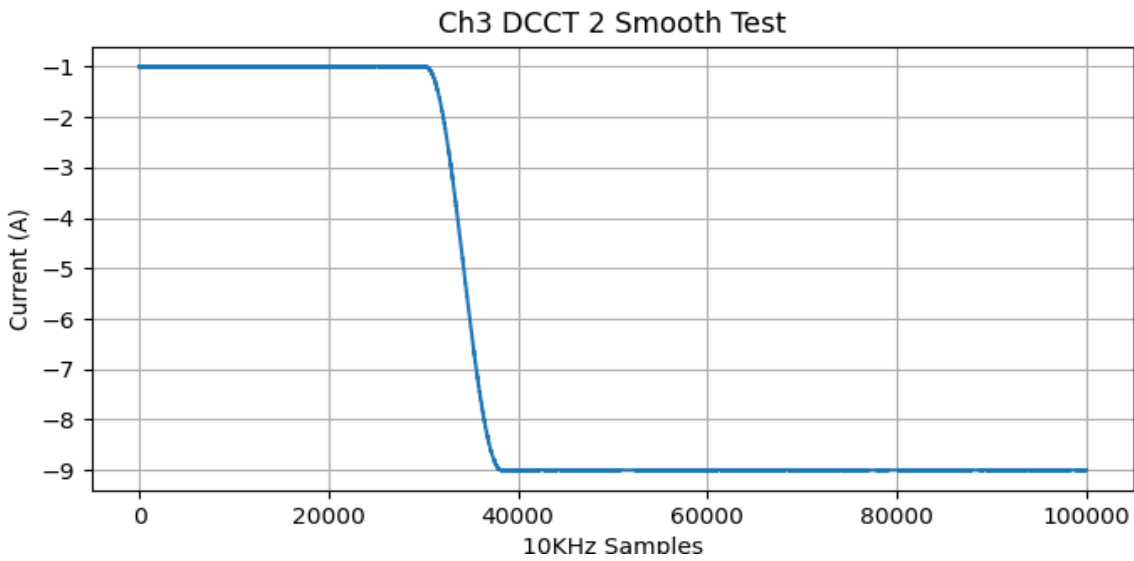
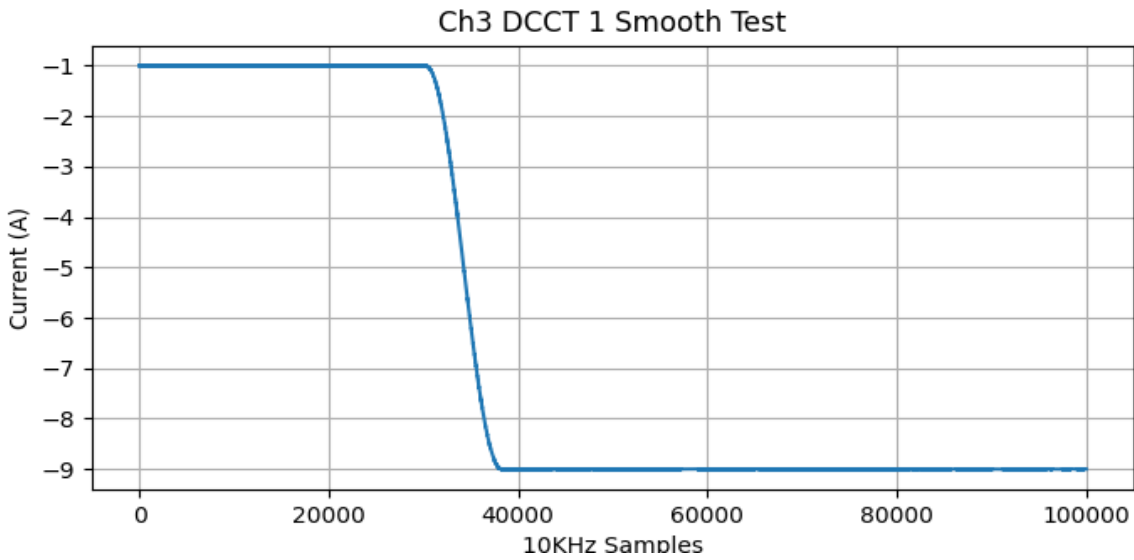
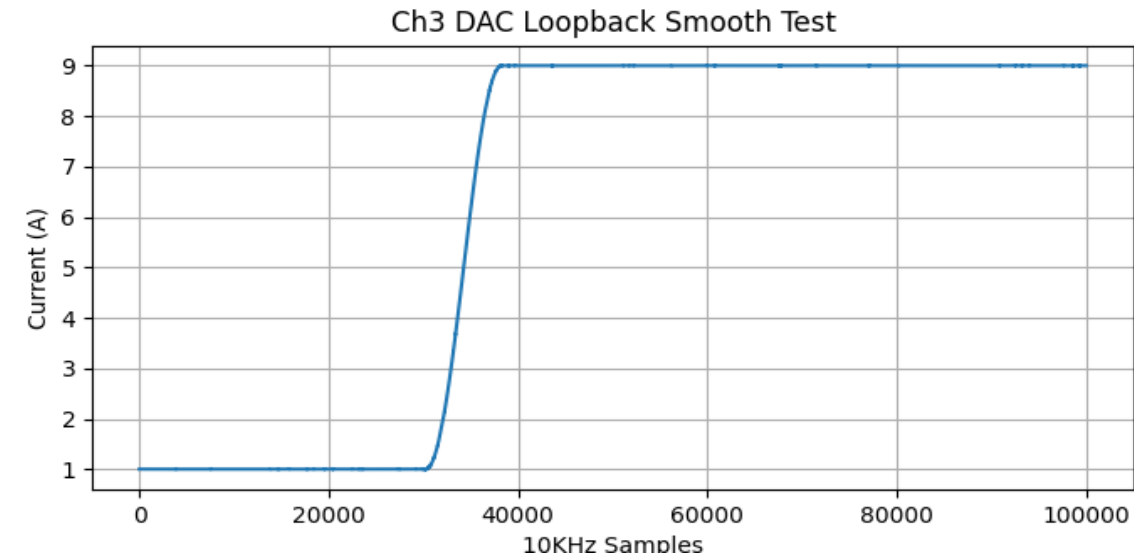
Jump Test Results: CH3



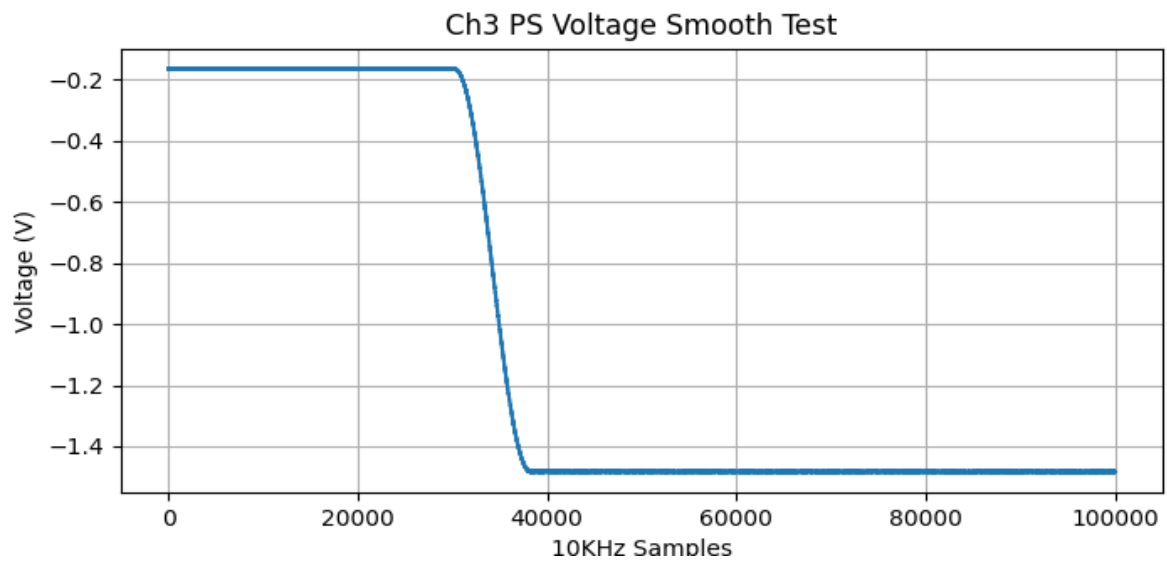
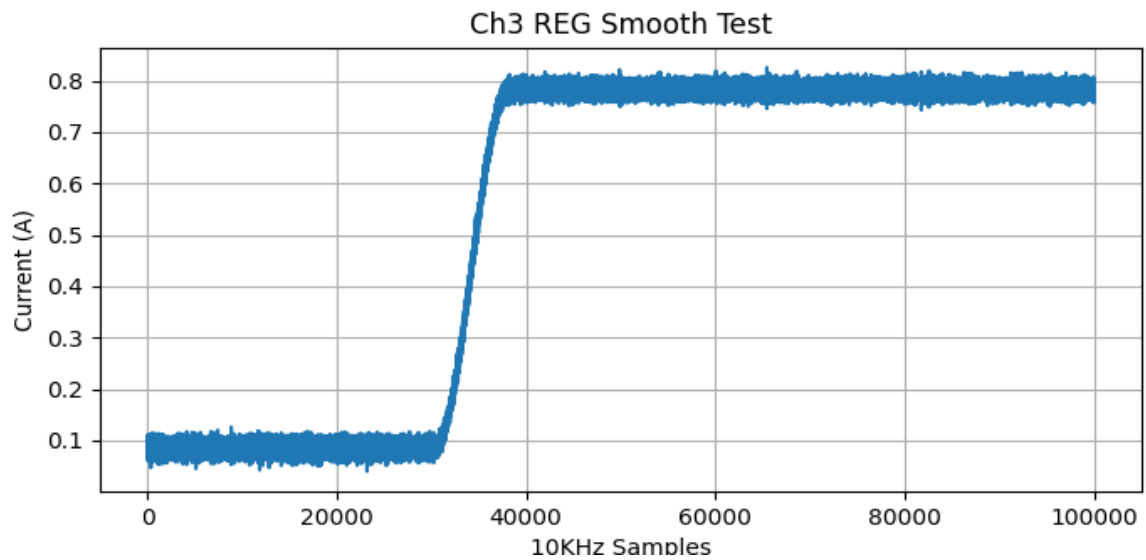
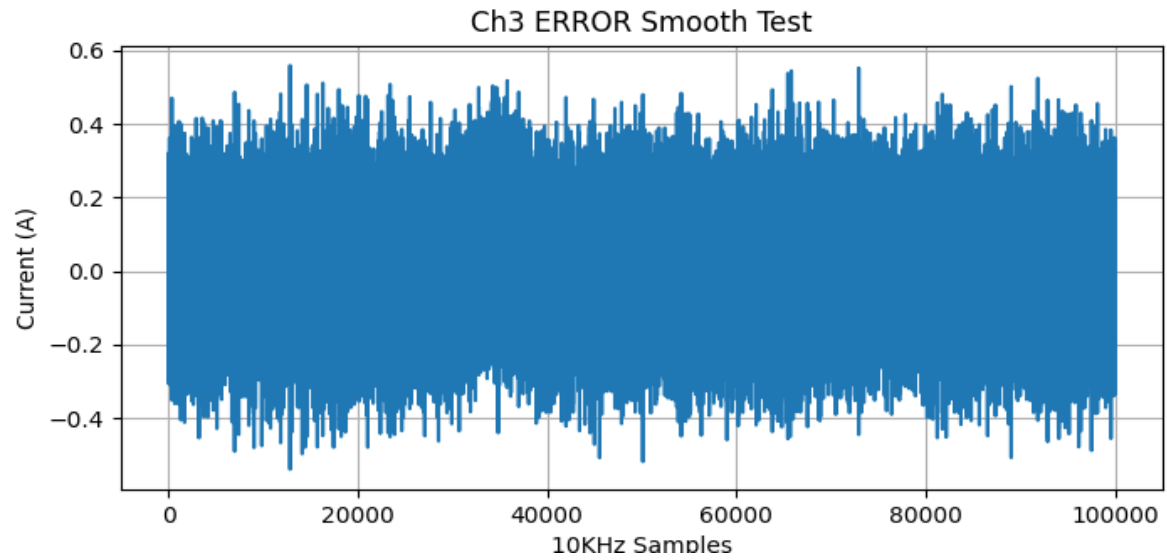
Jump Test Results: CH3



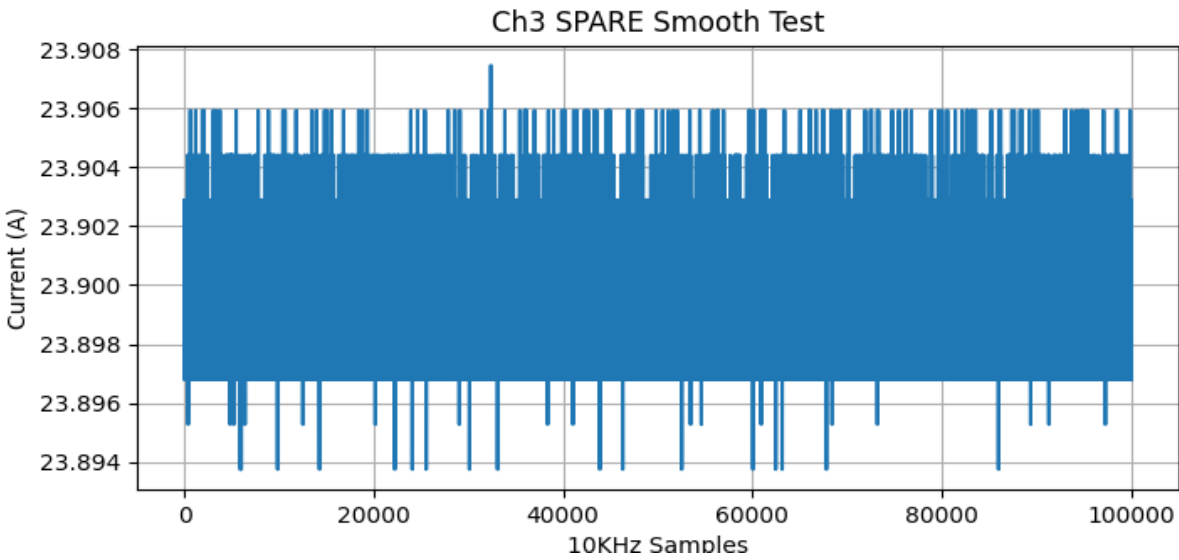
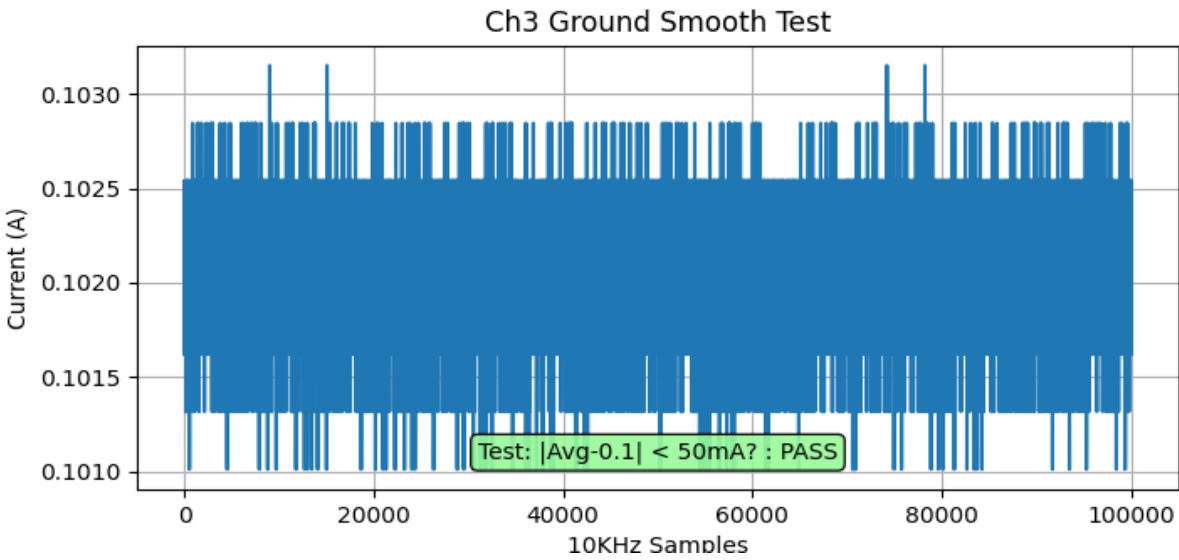
Smooth Test Results: CH3



Smooth Test Results: CH3



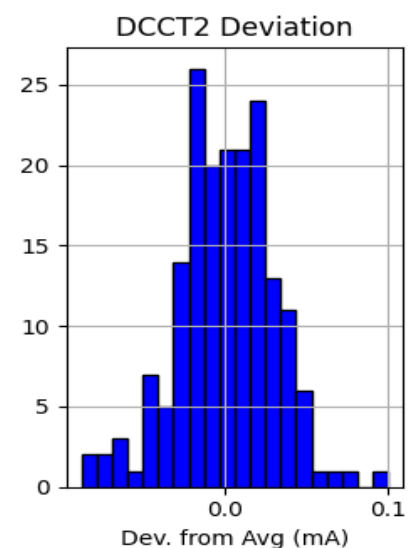
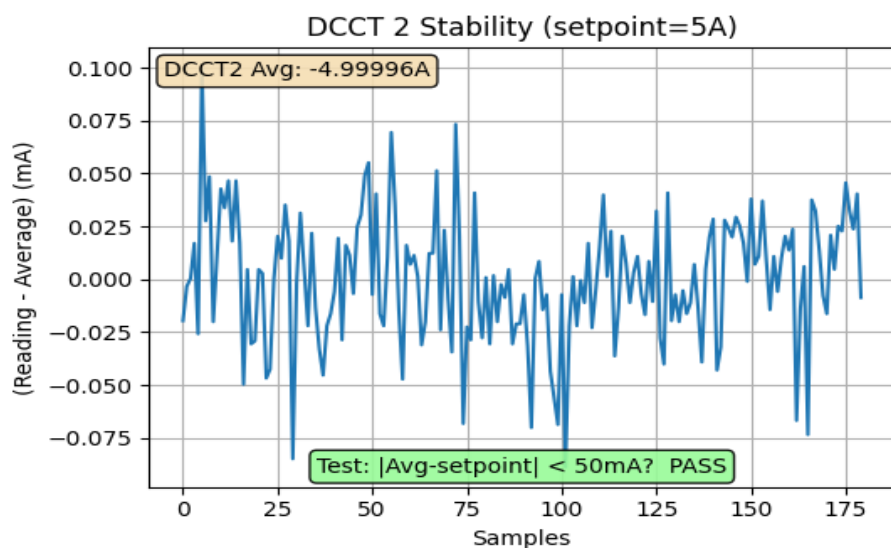
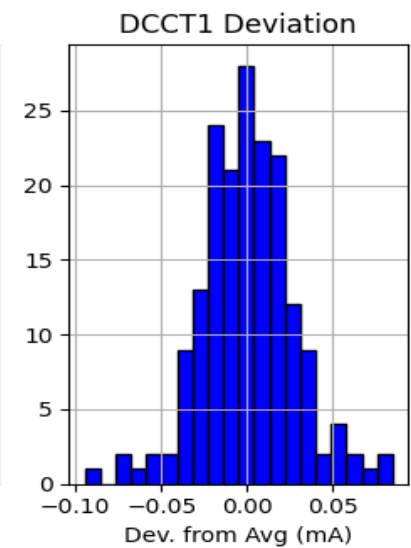
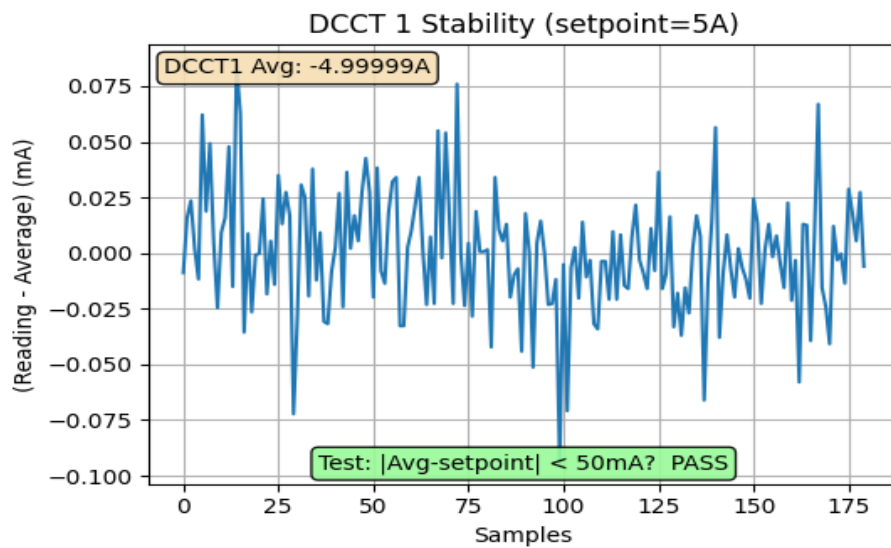
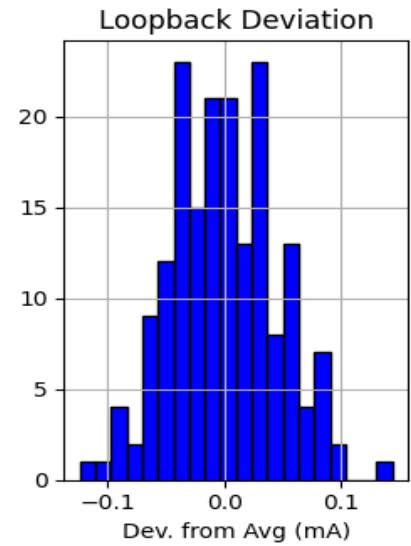
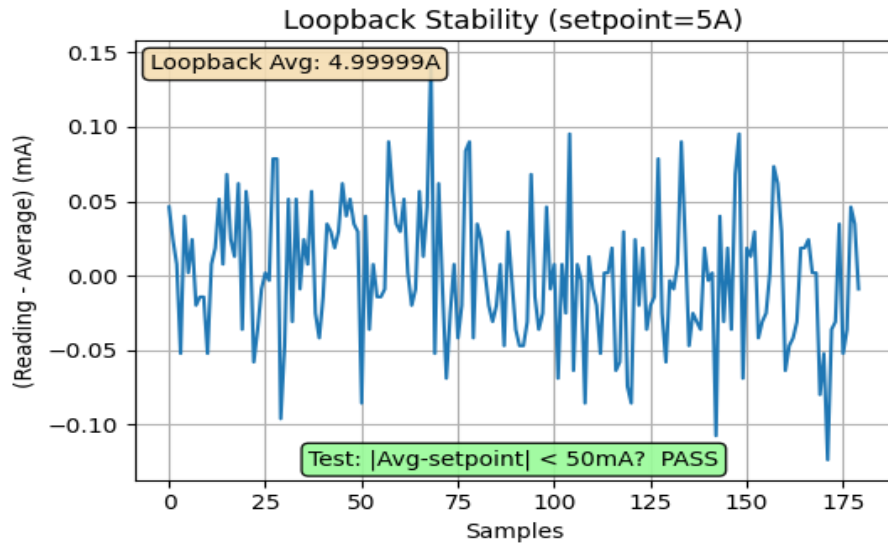
Smooth Test Results: CH3



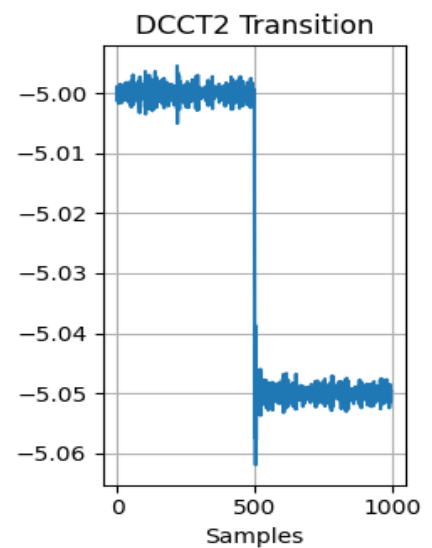
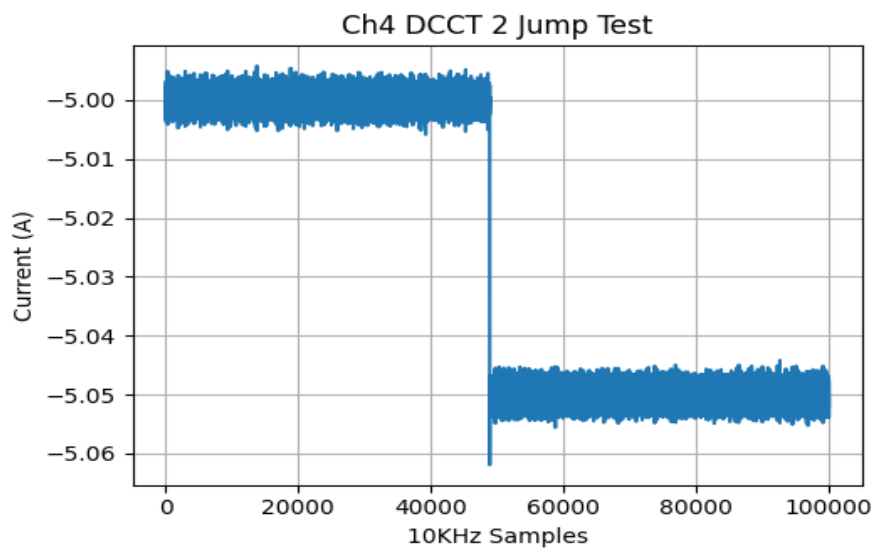
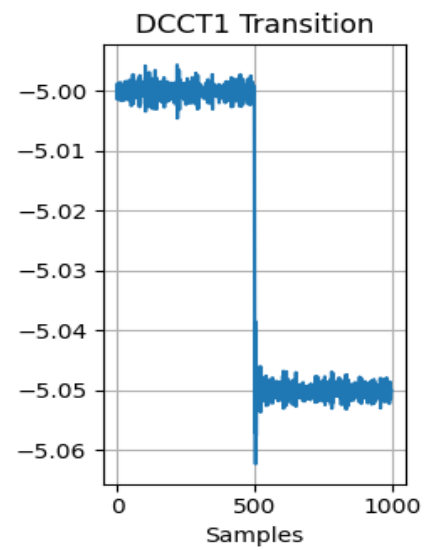
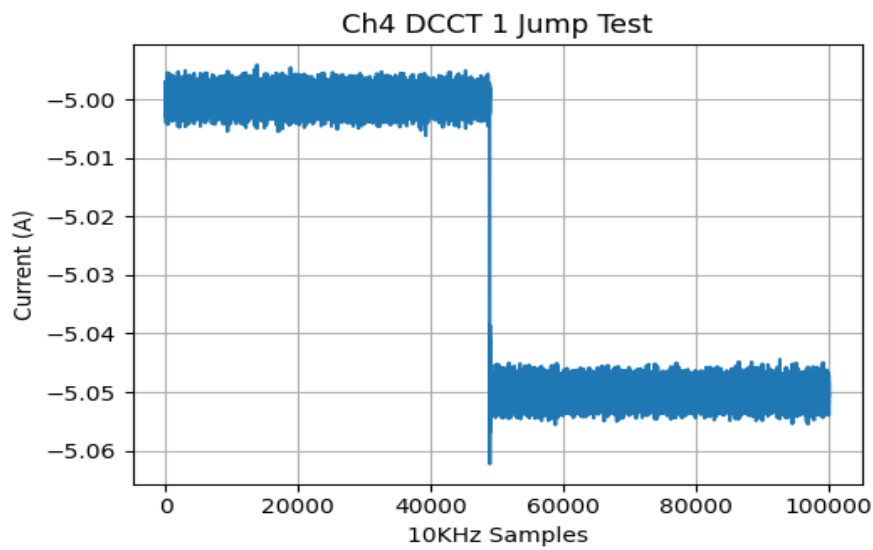
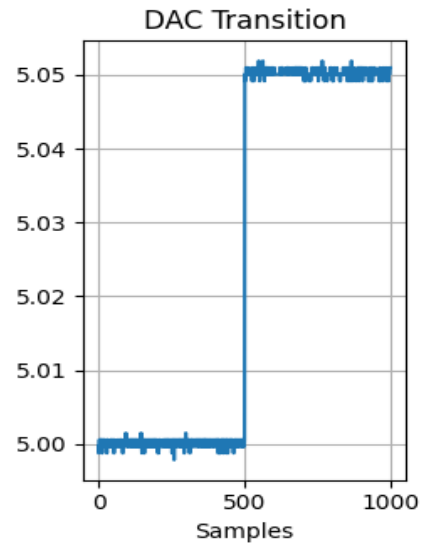
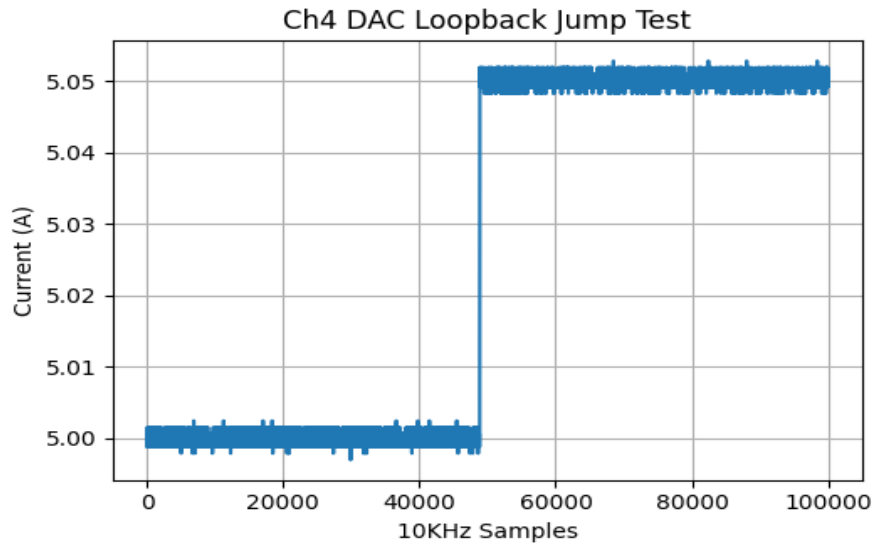
Channel 4

ATE Fault Tests for Channel 4	
Fault #1 Generated and Cleared	PASS
Fault #2 Generated and Cleared	PASS
Fault SPARE Generated and Cleared	PASS
Fault DCCT Generated and Cleared	PASS

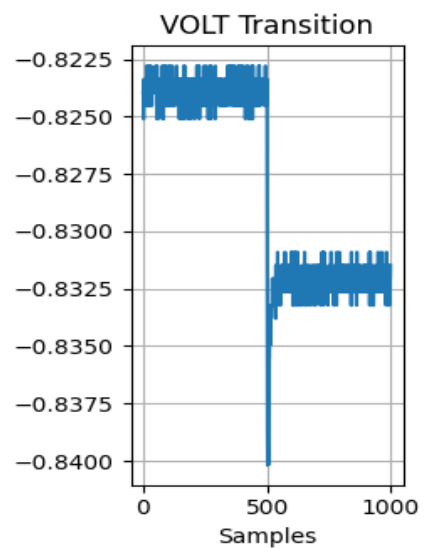
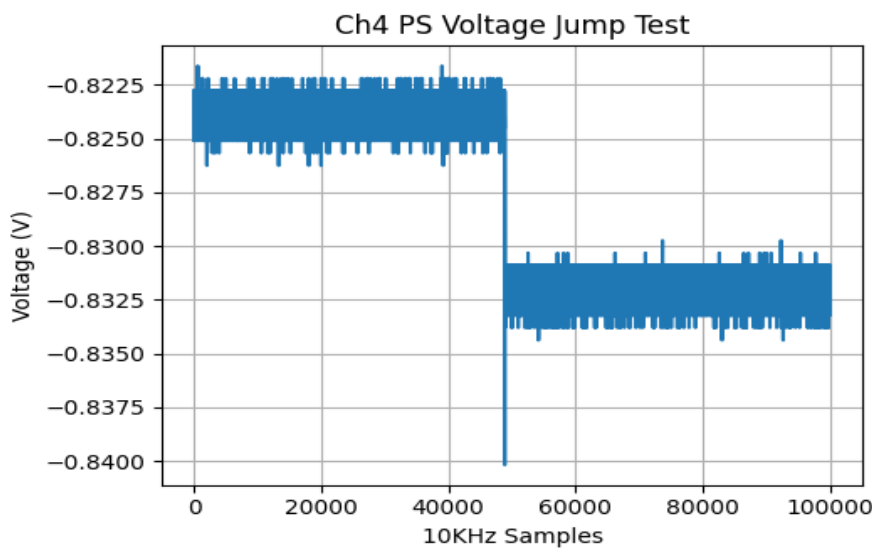
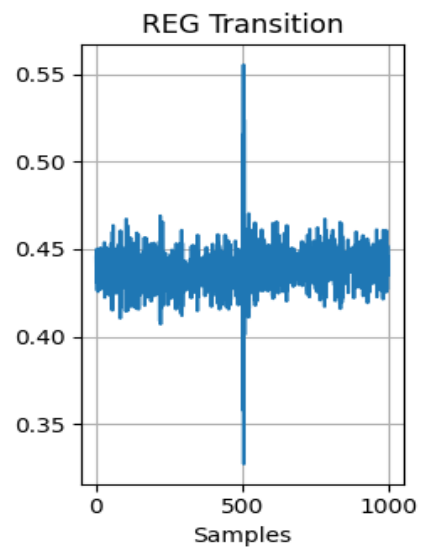
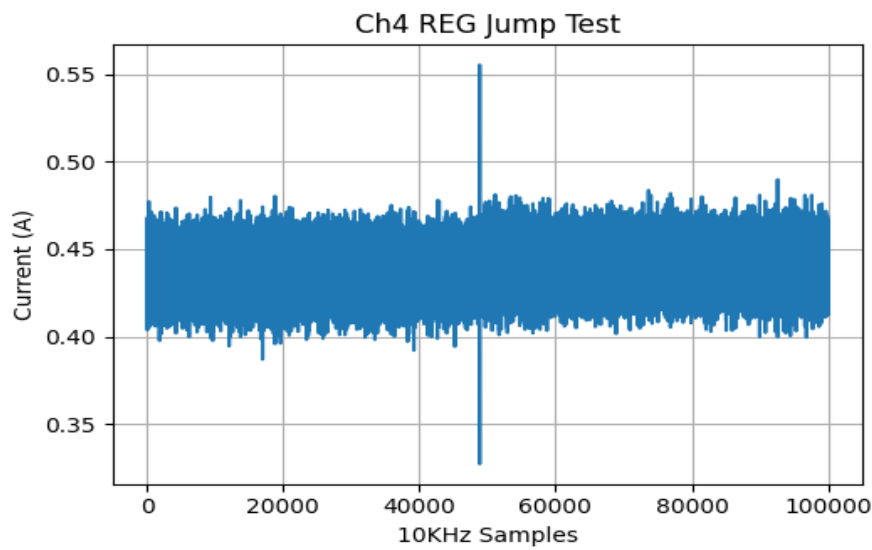
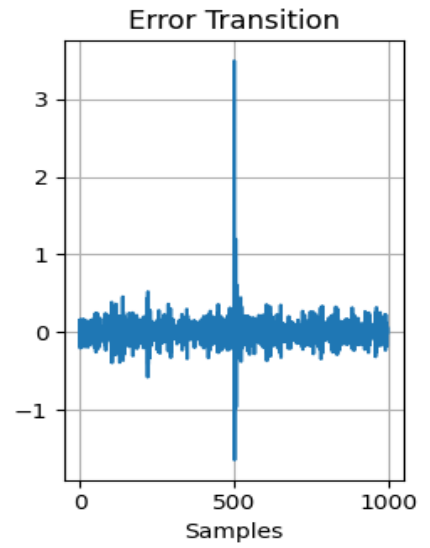
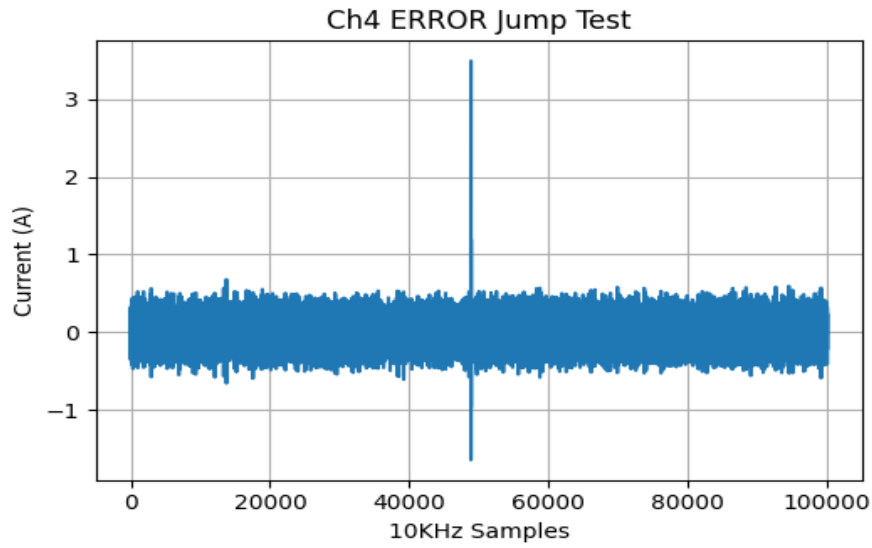
Power Supply Regulation for Channel 4:



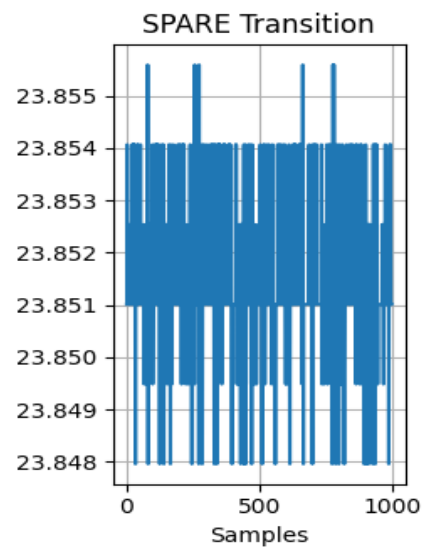
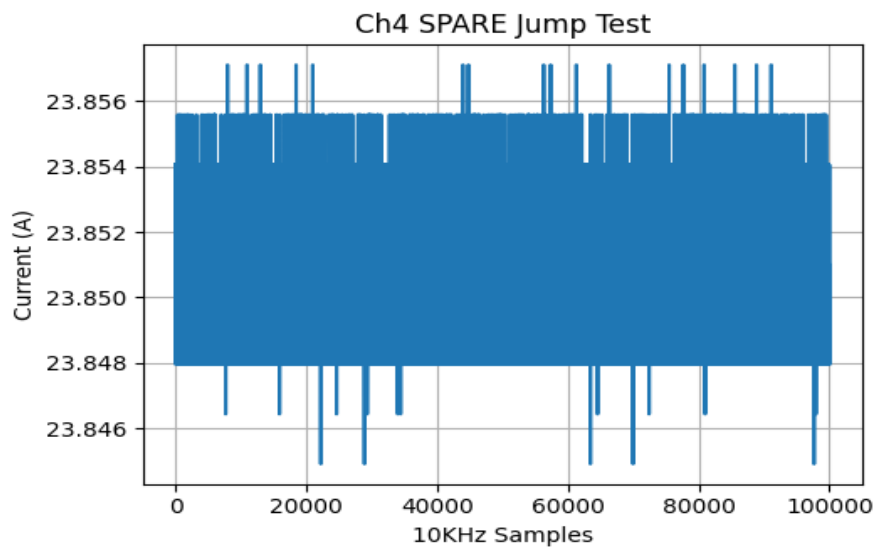
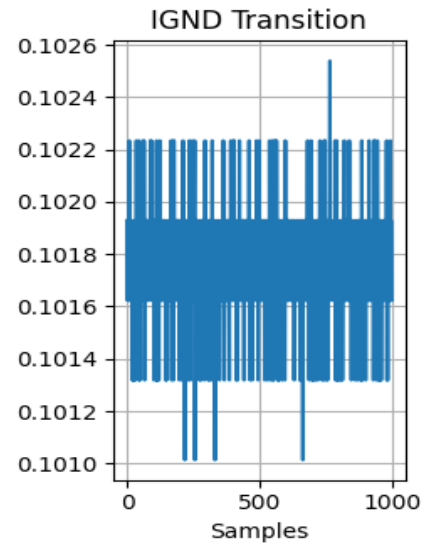
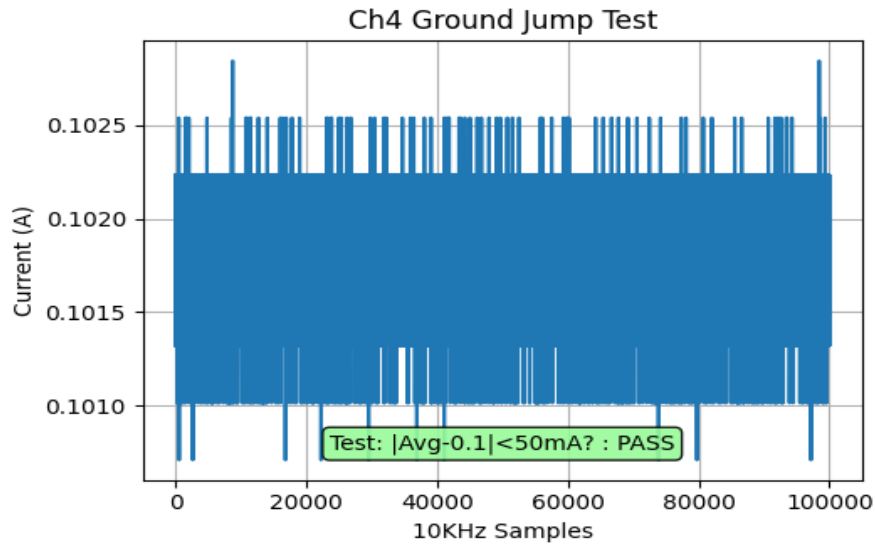
Jump Test Results: CH4



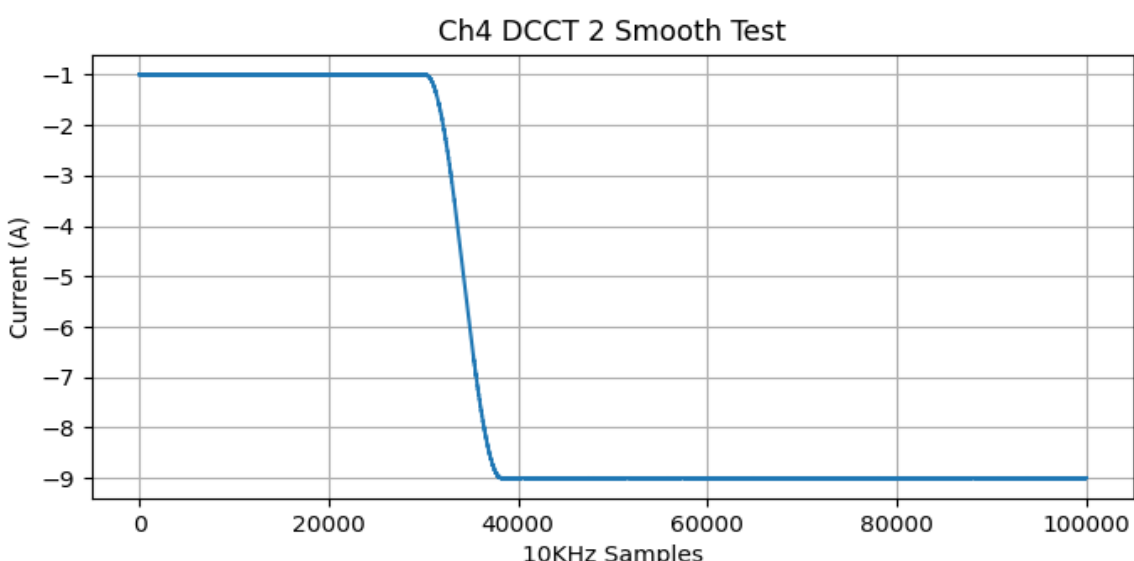
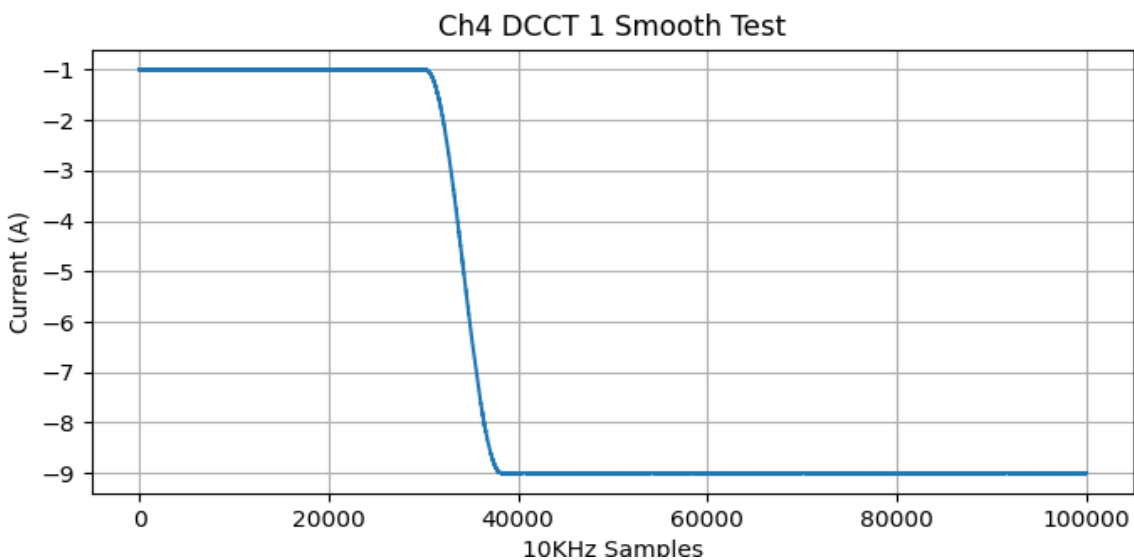
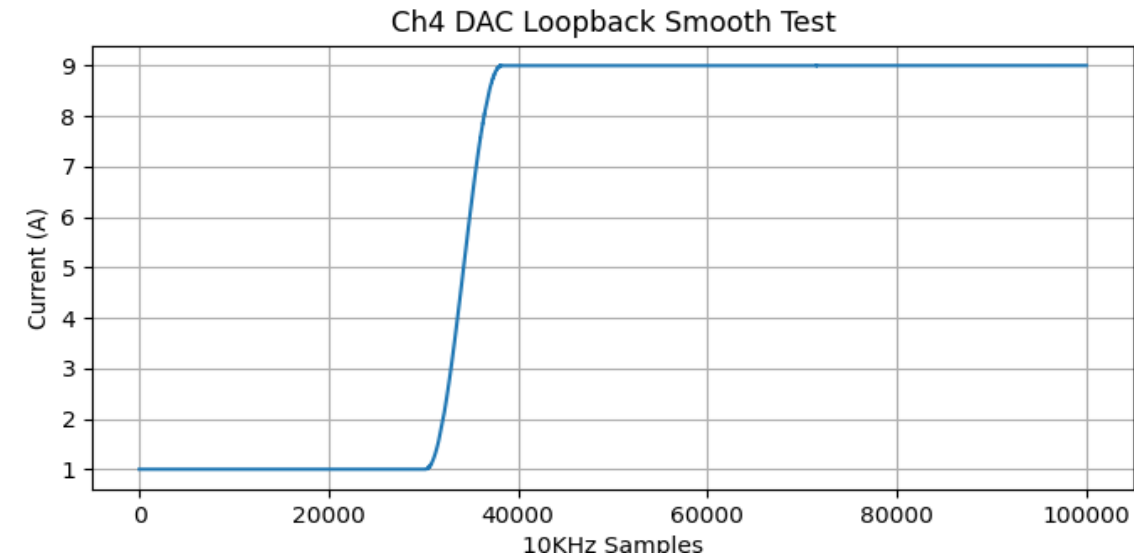
Jump Test Results: CH4



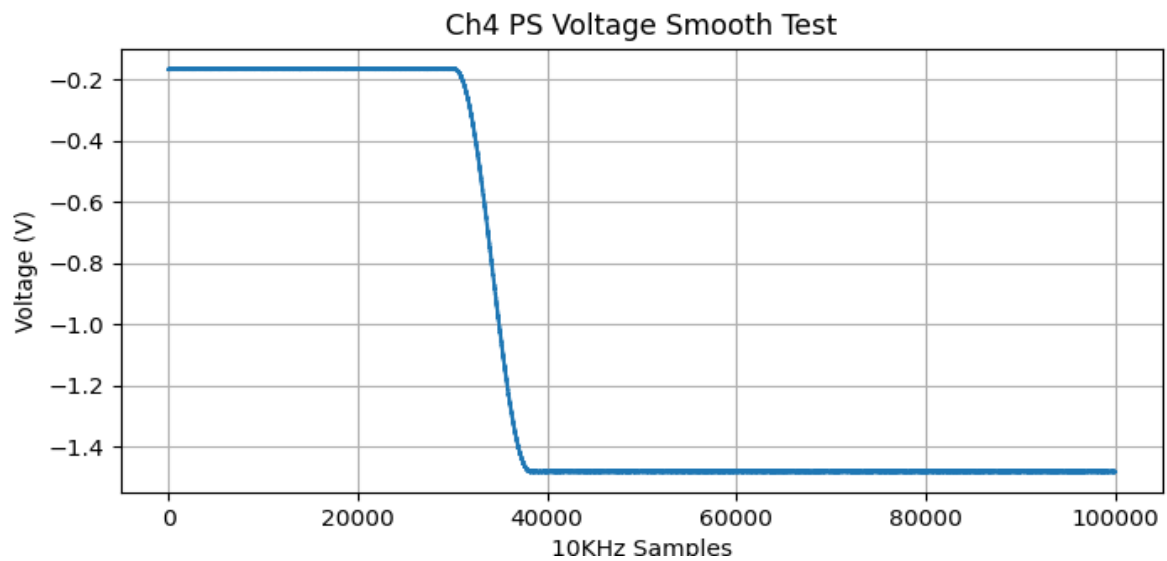
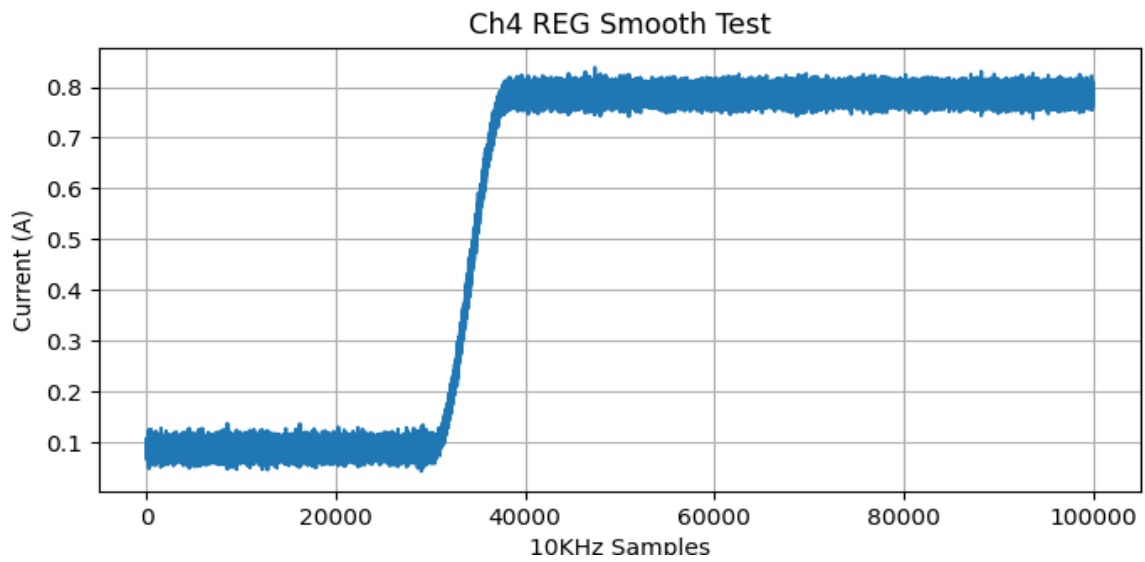
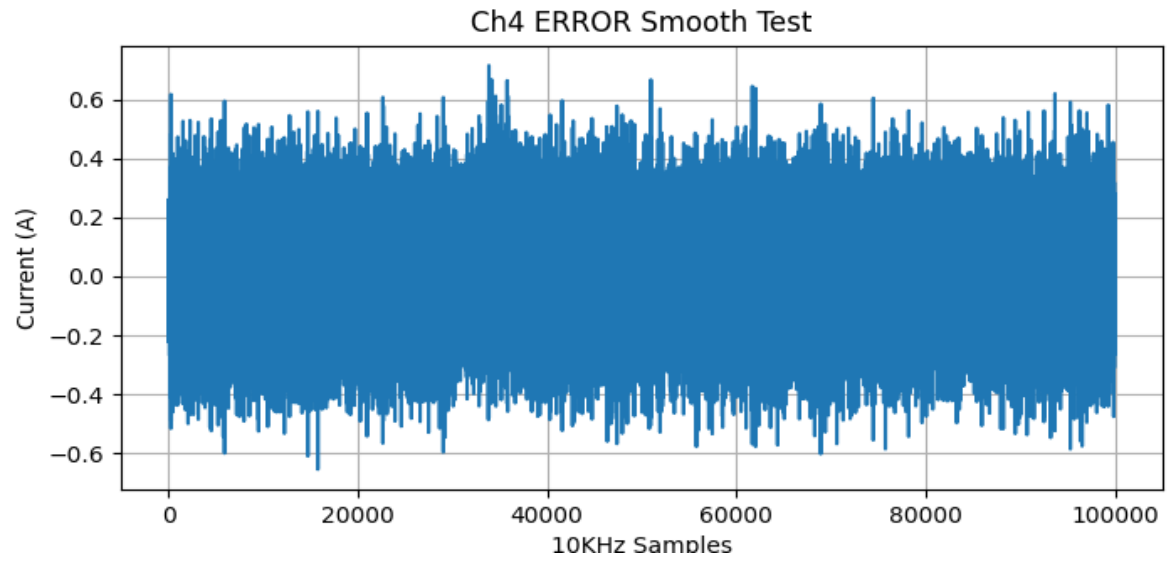
Jump Test Results: CH4



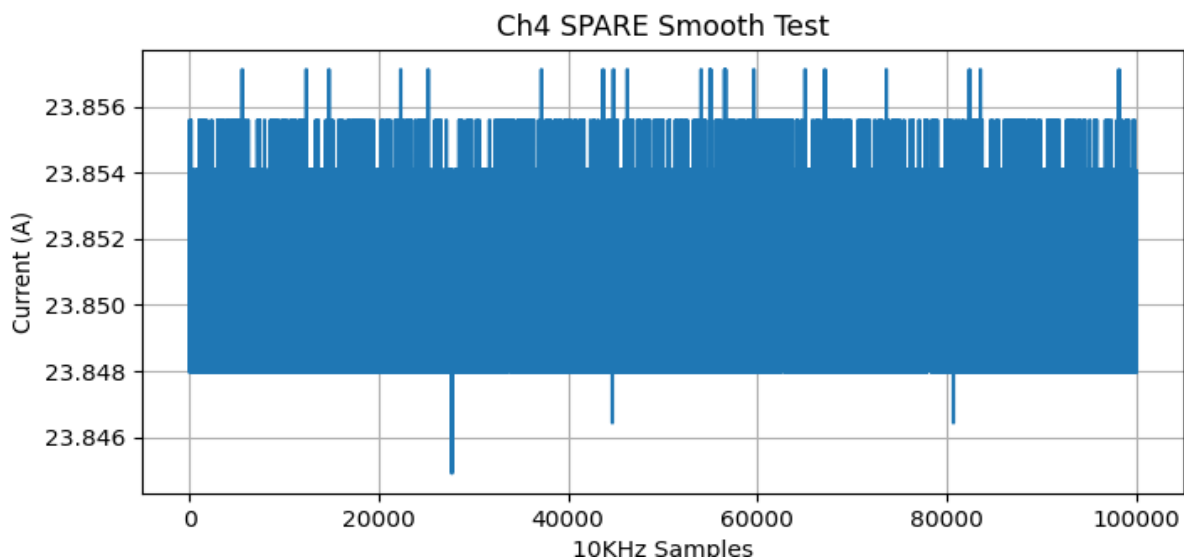
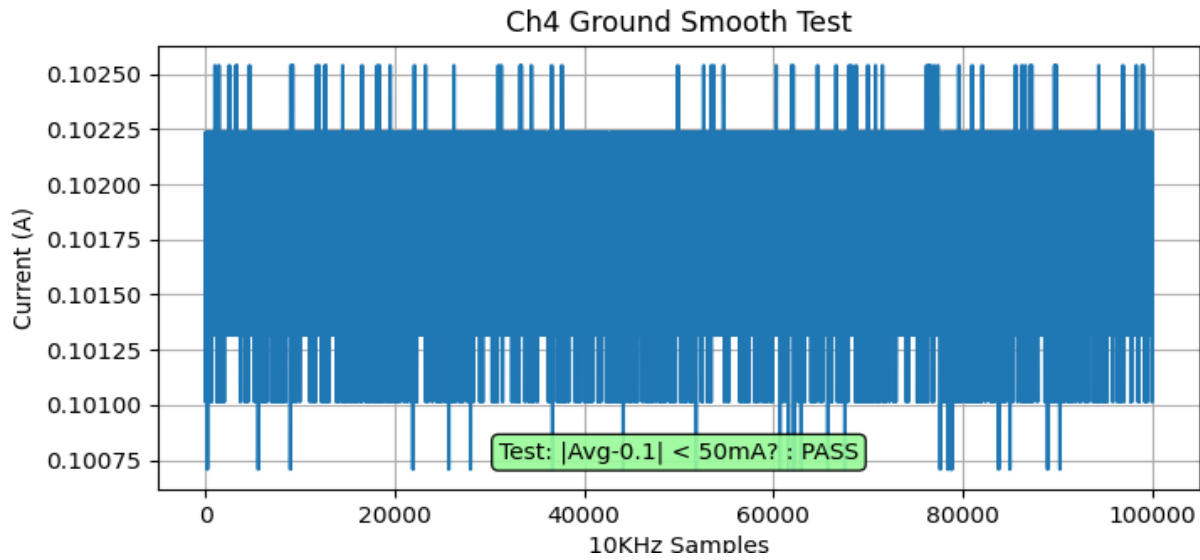
Smooth Test Results: CH4



Smooth Test Results: CH4



Smooth Test Results: CH4



FOFB TX test (Bandwidth-Mode = Fast)

PV	Value	Pass?
lab{2}Chan1:DAC-I	11.4988431930542	PASS
lab{2}Chan2:DAC-I	11.50019645690918	PASS
lab{2}Chan3:DAC-I	11.500794410705566	PASS
lab{2}Chan4:DAC-I	11.499785423278809	PASS

UDP RX Packet Test

```

23:18:00.587370 IP (tos 0x0, ttl 64, id 57082, offset 0, flags [DF], proto UDP (17), length 64)
  10.69.26.1.50654 > 10.69.26.55.12345: [udp sum ok] UDP, length 36
23:18:00.587371 IP (tos 0x0, ttl 64, id 57083, offset 0, flags [DF], proto UDP (17), length 64)
  10.69.26.1.49639 > 10.69.26.55.12345: [udp sum ok] UDP, length 36
23:18:00.598781 IP (tos 0x0, ttl 64, id 57084, offset 0, flags [DF], proto UDP (17), length 64)
  10.69.26.1.55376 > 10.69.26.55.12345: [udp sum ok] UDP, length 36
23:18:00.598781 IP (tos 0x0, ttl 64, id 57085, offset 0, flags [DF], proto UDP (17), length 64)
  10.69.26.1.41305 > 10.69.26.55.12345: [udp sum ok] UDP, length 36
23:18:00.610654 IP (tos 0x0, ttl 64, id 57087, offset 0, flags [DF], proto UDP (17), length 64)
  10.69.26.1.55951 > 10.69.26.55.12345: [udp sum ok] UDP, length 36
23:18:00.610655 IP (tos 0x0, ttl 64, id 57088, offset 0, flags [DF], proto UDP (17), length 64)
  10.69.26.1.60133 > 10.69.26.55.12345: [udp sum ok] UDP, length 36
23:18:00.622344 IP (tos 0x0, ttl 64, id 57093, offset 0, flags [DF], proto UDP (17), length 64)
  10.69.26.1.55128 > 10.69.26.55.12345: [udp sum ok] UDP, length 36
23:18:00.622344 IP (tos 0x0, ttl 64, id 57094, offset 0, flags [DF], proto UDP (17), length 64)
  10.69.26.1.42697 > 10.69.26.55.12345: [udp sum ok] UDP, length 36
23:18:00.633813 IP (tos 0x0, ttl 64, id 57095, offset 0, flags [DF], proto UDP (17), length 64)
  10.69.26.1.36116 > 10.69.26.55.12345: [udp sum ok] UDP, length 36
23:18:00.633814 IP (tos 0x0, ttl 64, id 57096, offset 0, flags [DF], proto UDP (17), length 64)
  10.69.26.1.36741 > 10.69.26.55.12345: [udp sum ok] UDP, length 36
23:18:00.645485 IP (tos 0x0, ttl 64, id 57103, offset 0, flags [DF], proto UDP (17), length 64)
  10.69.26.1.54688 > 10.69.26.55.12345: [udp sum ok] UDP, length 36
23:18:00.645486 IP (tos 0x0, ttl 64, id 57104, offset 0, flags [DF], proto UDP (17), length 64)
  10.69.26.1.37310 > 10.69.26.55.12345: [udp sum ok] UDP, length 36
23:18:00.656951 IP (tos 0x0, ttl 64, id 57110, offset 0, flags [DF], proto UDP (17), length 64)
  10.69.26.1.50865 > 10.69.26.55.12345: [udp sum ok] UDP, length 36
23:18:00.656951 IP (tos 0x0, ttl 64, id 57111, offset 0, flags [DF], proto UDP (17), length 64)
  10.69.26.1.39005 > 10.69.26.55.12345: [udp sum ok] UDP, length 36
23:18:00.668238 IP (tos 0x0, ttl 64, id 57120, offset 0, flags [DF], proto UDP (17), length 64)
  10.69.26.1.34493 > 10.69.26.55.12345: [udp sum ok] UDP, length 36
23:18:00.668239 IP (tos 0x0, ttl 64, id 57121, offset 0, flags [DF], proto UDP (17), length 64)
  10.69.26.1.53892 > 10.69.26.55.12345: [udp sum ok] UDP, length 36
23:18:00.679692 IP (tos 0x0, ttl 64, id 57127, offset 0, flags [DF], proto UDP (17), length 64)
  10.69.26.1.37693 > 10.69.26.55.12345: [udp sum ok] UDP, length 36
23:18:00.679693 IP (tos 0x0, ttl 64, id 57128, offset 0, flags [DF], proto UDP (17), length 64)
  10.69.26.1.47450 > 10.69.26.55.12345: [udp sum ok] UDP, length 36
23:18:00.690998 IP (tos 0x0, ttl 64, id 57140, offset 0, flags [DF], proto UDP (17), length 64)
  10.69.26.1.35218 > 10.69.26.55.12345: [udp sum ok] UDP, length 36
23:18:00.690998 IP (tos 0x0, ttl 64, id 57141, offset 0, flags [DF], proto UDP (17), length 64)
  10.69.26.1.58880 > 10.69.26.55.12345: [udp sum ok] UDP, length 36

```

(Full tcpdump log saved to /home/pstester/NSLS-II_PSCs/NEXT-II/NSLS2_zPSC_Calibration_and_Test/Test_Data/Raw_Logs/4ch_HSF_SN0001_RawData_04-30-26_22-45/tcpdump_full.log)

Test	Result
UDP RX FOFB	PASS